NAME : Shri Harshini Devi . A

REG.NO : 192524104

Dept. Name : B.Tech-AIDS

Course Code/Course Name: DSA0602 /Data Handling and Visualization

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LIST OF EXPERIMENTS

Set 21

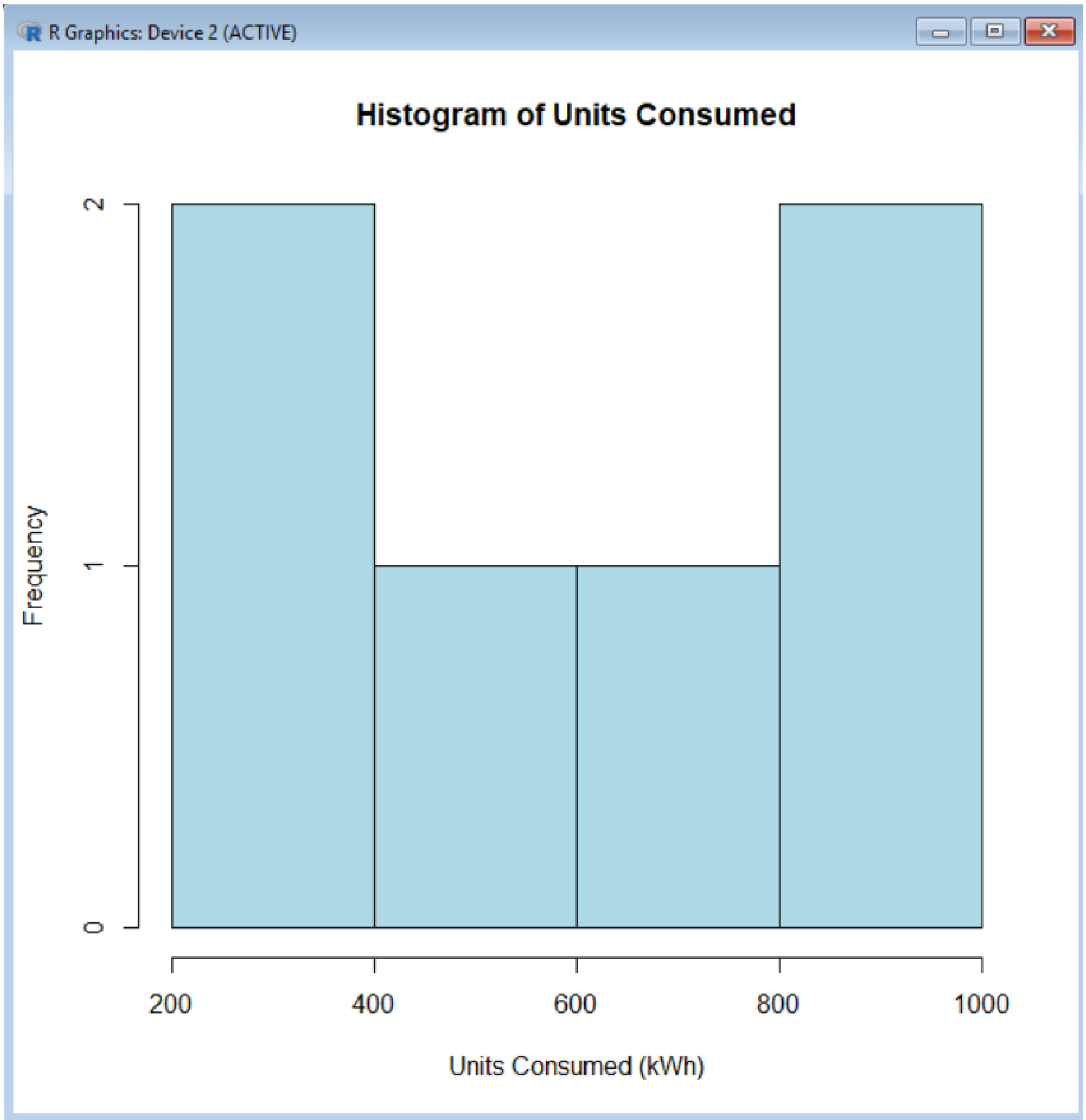
Dataset: Energy_Consumption_Data.csv

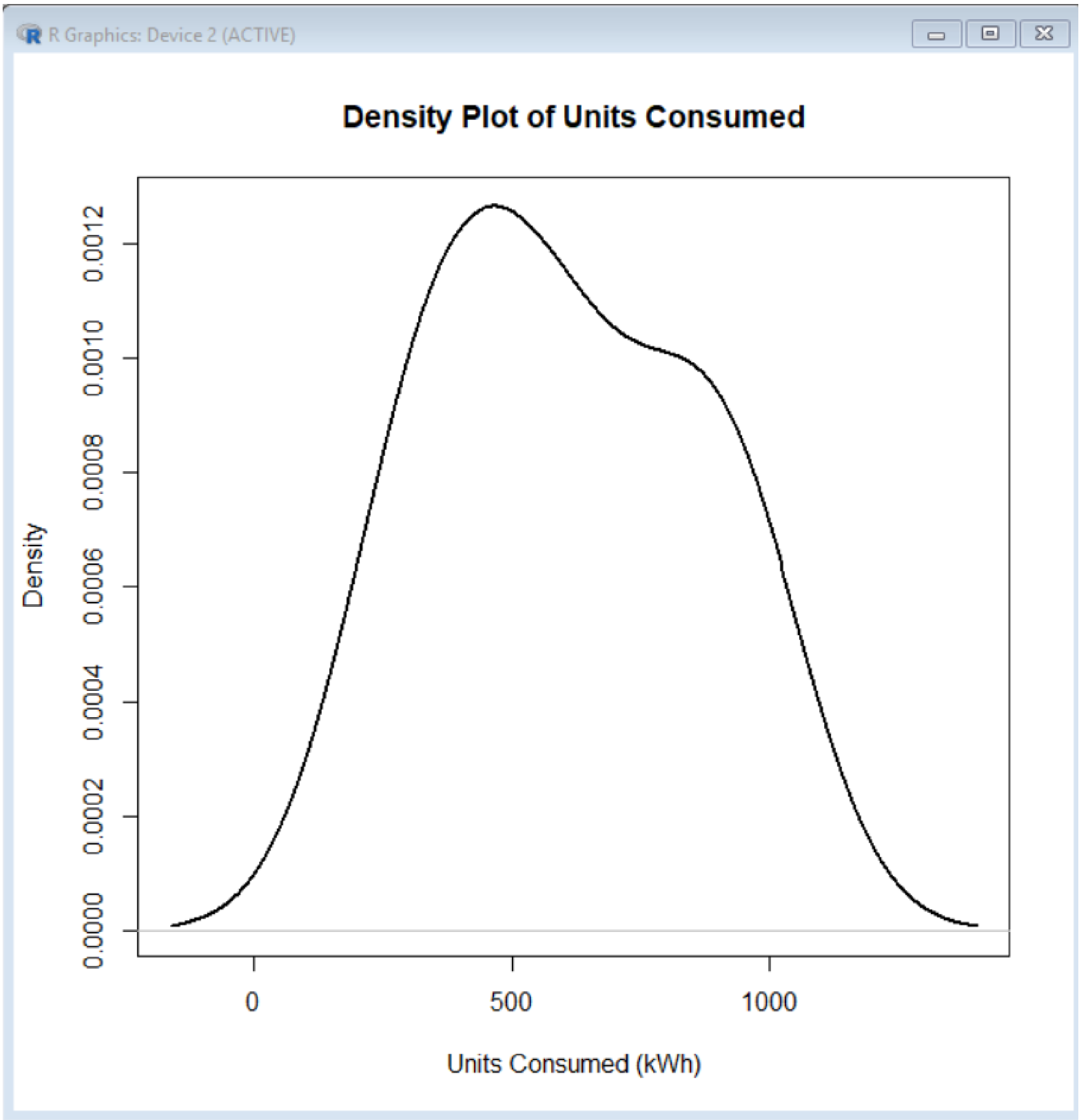
Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

1. **Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.**

```
# Import dataset

data <- read.csv("Energy_Consumption_Data.csv")
```

```
# View dataset
print(data)

# Histogram for Units_Consumed
hist(data$Units_Consumed,
    main = "Histogram of Energy Consumption",
    xlab = "Units Consumed (kWh)",
    col = "lightblue",
    border = "black")

# Density plot
plot(density(data$Units_Consumed),
    main = "Density Plot of Energy Consumption",
    xlab = "Units Consumed (kWh)",
    col = "red",
    lwd = 2)
```


R Graphics: Device 2 (ACTIVE)
Histogram of Units Consumed
Frequency
0
1
2
200
400
600
800
1000
Units Consumed (kWh)

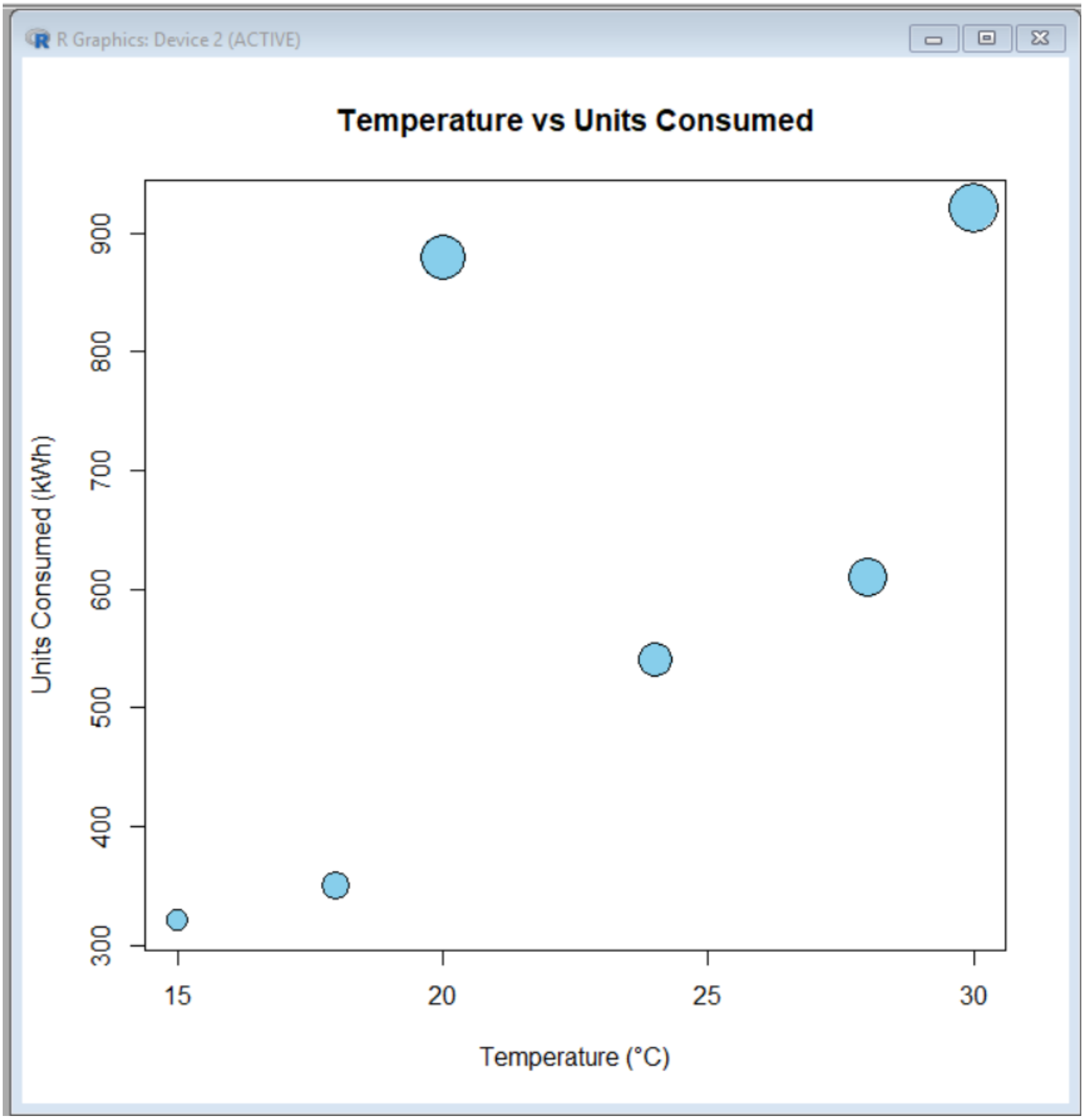


2. **Using R programming, create a scatter plot between Temperature and Units_Consumed.Represent Peak_Hours using bubble size. Apply transparency.**

```
plot(energy$Temperature, energy$Units_Consumed,

    main = "Temperature vs Units Consumed",

    xlab = "Temperature (°C)",

    ylab = "Units Consumed (kWh)",

    pch = 21,

    bg = "skyblue",
```

cex = energy$Peak_Hours / 2,

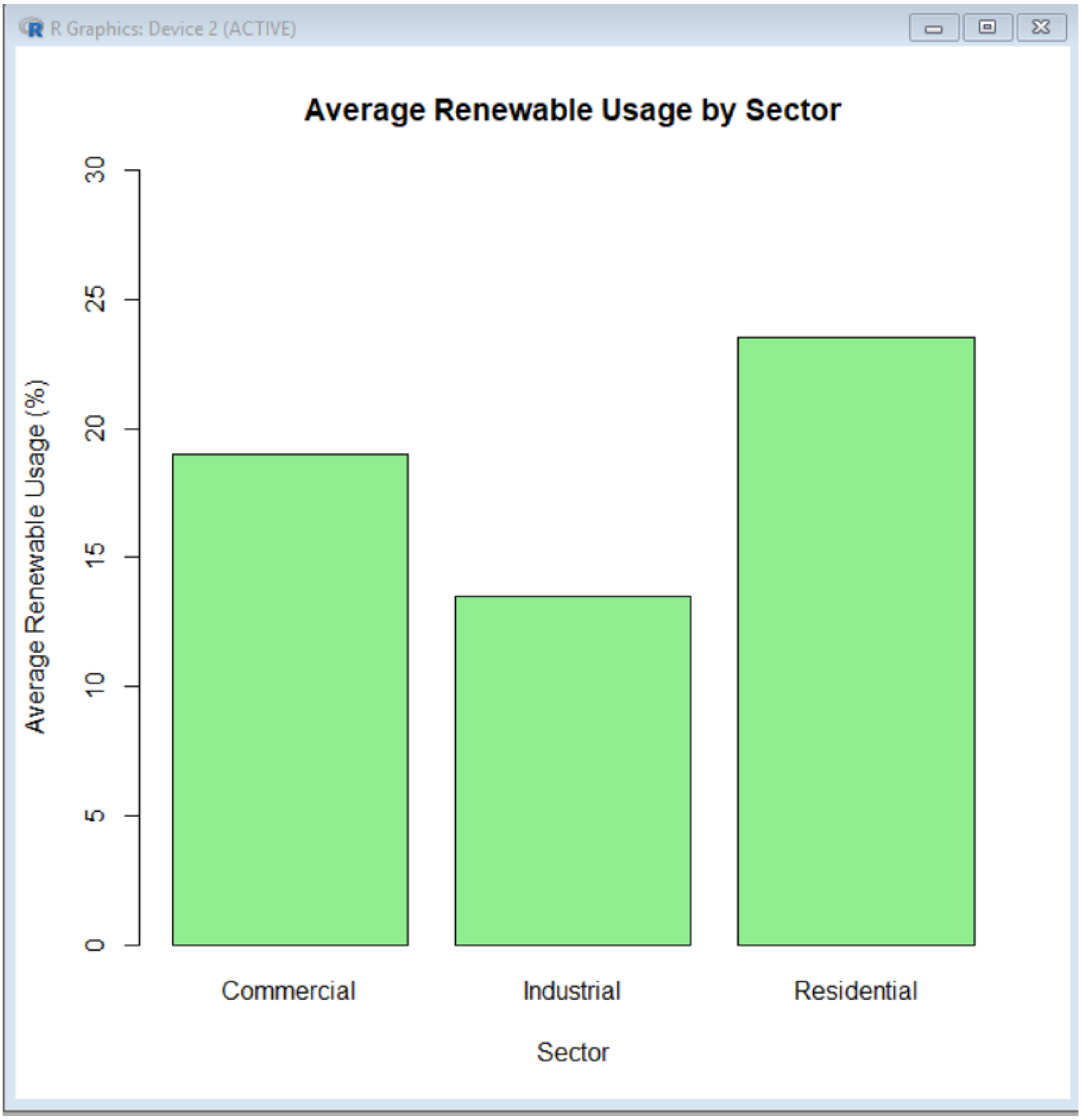
col = "black")



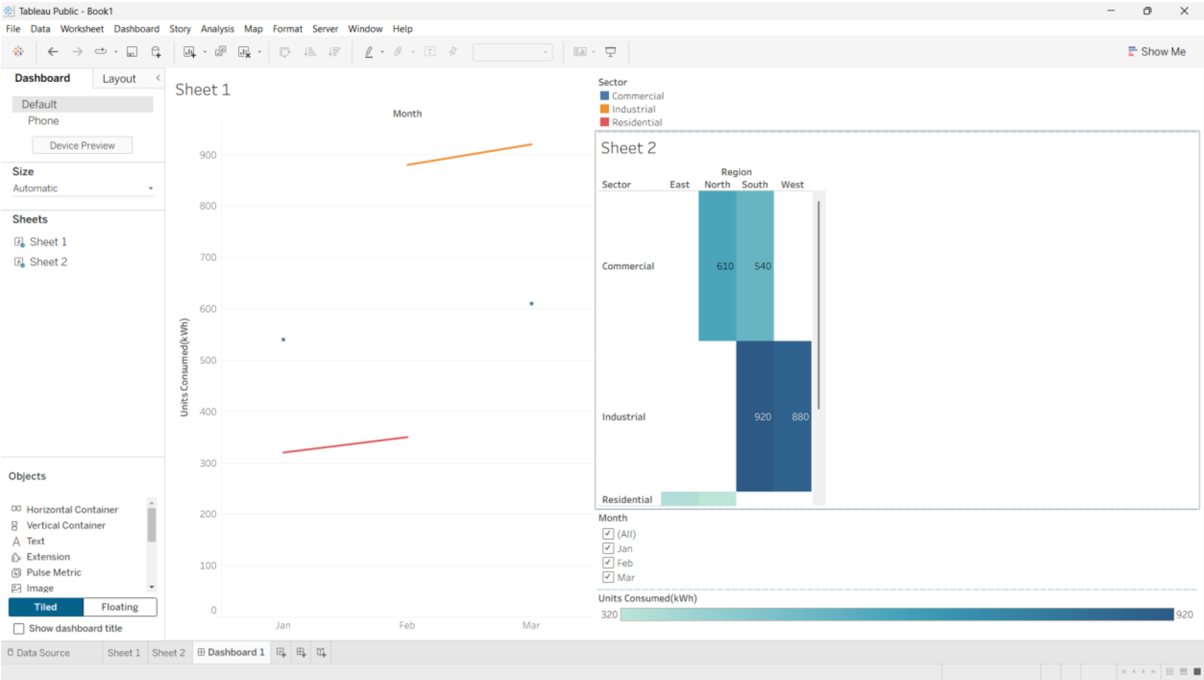
3. **Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.**

```
barplot(avg_renewable$Renewable_Usage,

    names.arg = avg_renewable$Sector,

    main = "Average Renewable Usage by Sector",

    xlab = "Sector",
```

```
ylab = "Average Renewable Usage (%)",
col = "lightgreen",
ylim = c(0, 30))
```



4. Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.



Set 22: Time Series Analysis

Dataset: Monthly Sales Data

Month	Sales (in $)
January	15000
February	18000
March	22000
April	20000

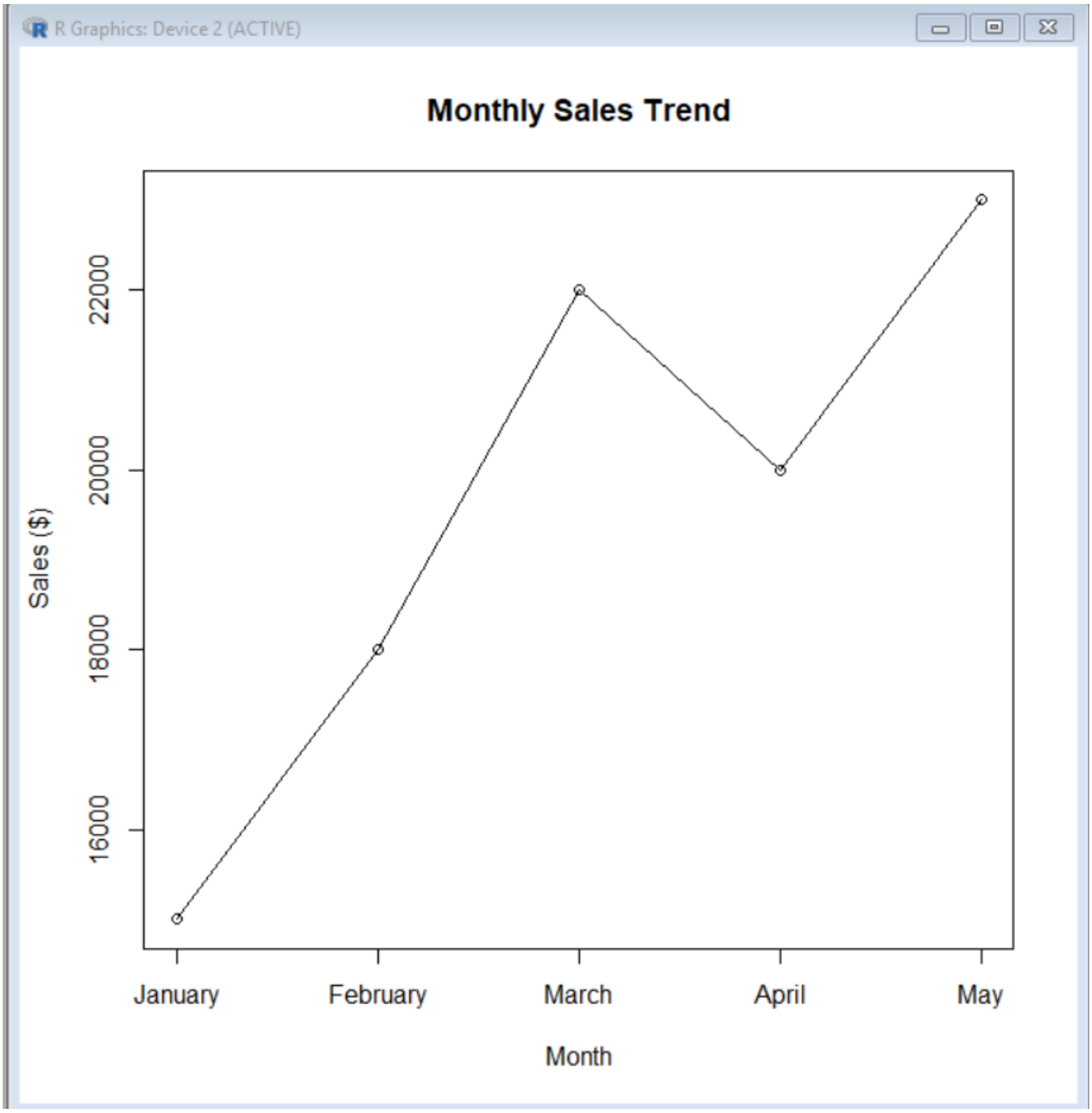
1. **In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.**

```
sales <- c(15000, 18000, 22000, 20000, 23000)

months <- c("January", "February", "March", "April", "May")

plot(sales,
    type = "o",
    main = "Monthly Sales Trend",
    xlab = "Month",
```

ylab = "Sales ($)",

xaxt = "n")

axis(1, at = 1:5, labels = months)

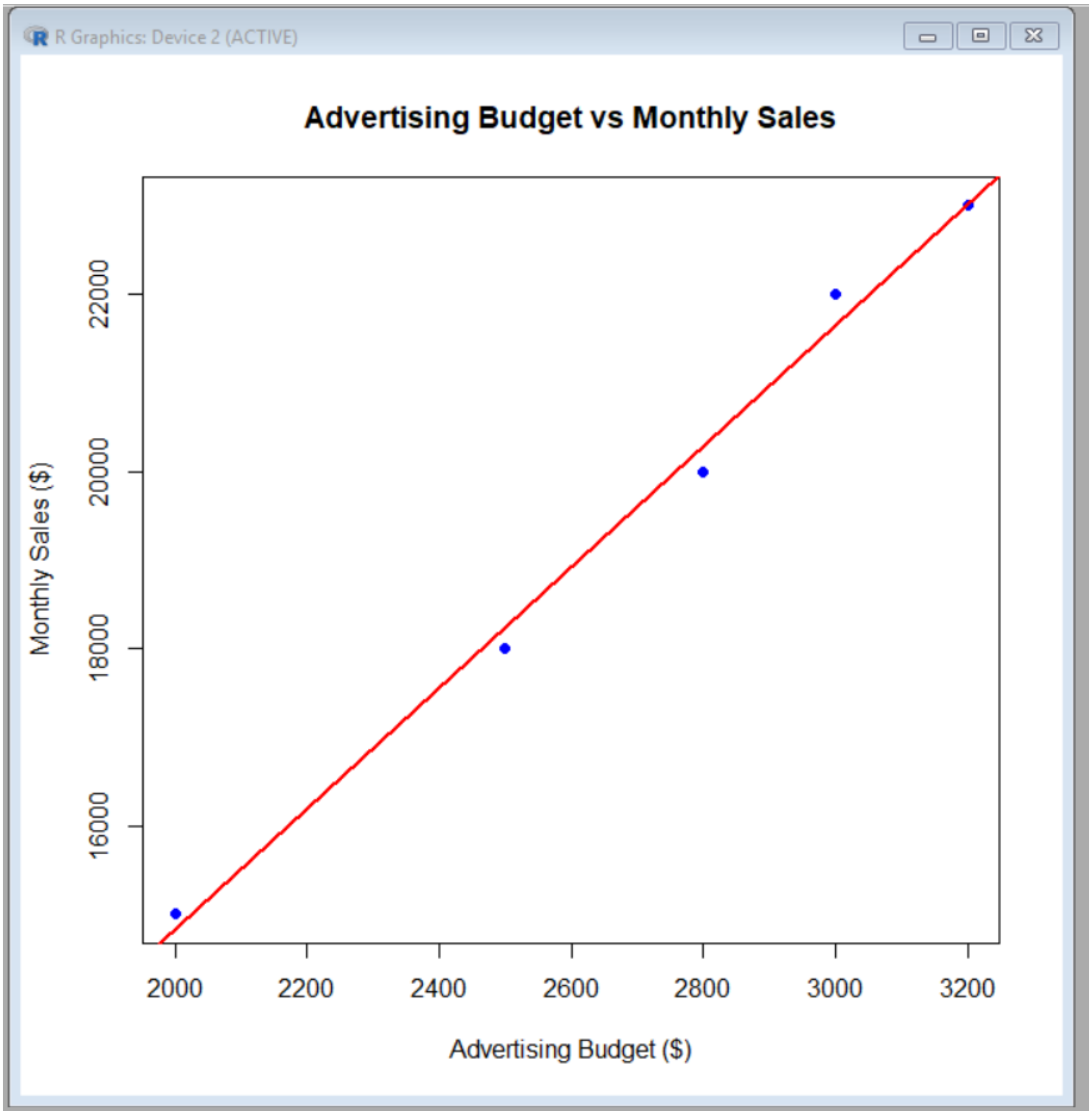


2. **In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.**

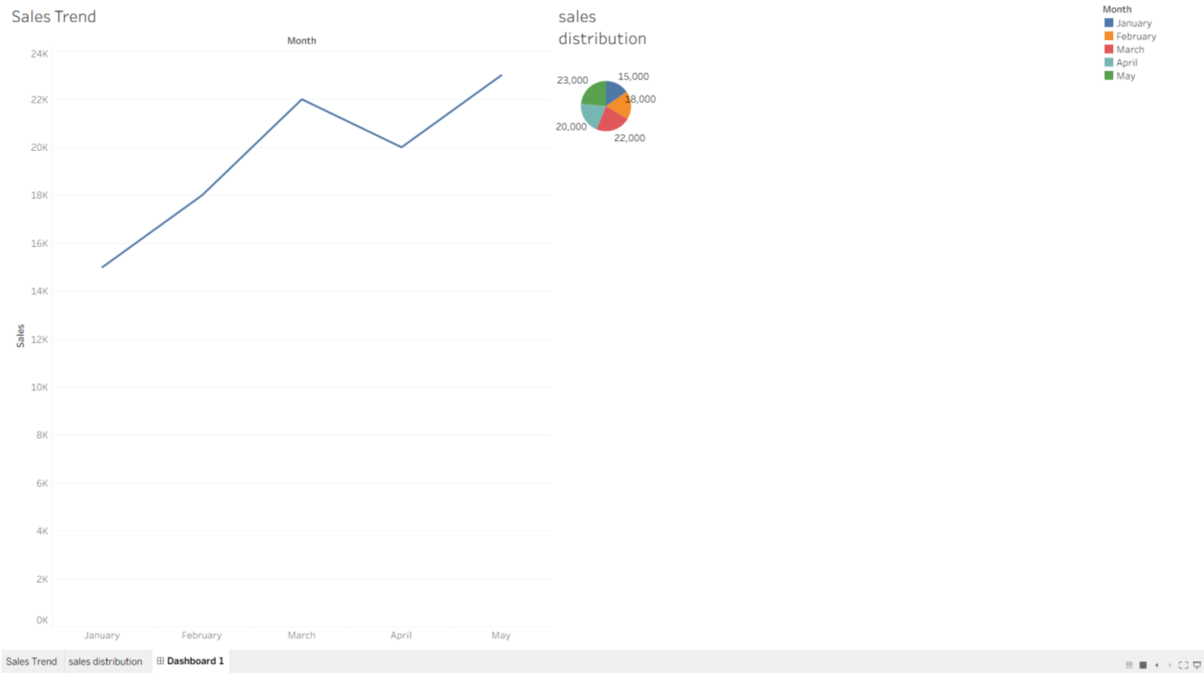
advertising <- c(2000, 2500, 3000, 2800, 3200)

sales <- c(15000, 18000, 22000, 20000, 23000)

```
plot(advertising, sales,
     main = "Advertising Budget vs Monthly Sales",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     pch = 19,
     col = "blue")

abline(lm(sales ~ advertising), col = "red", lwd = 2)
```

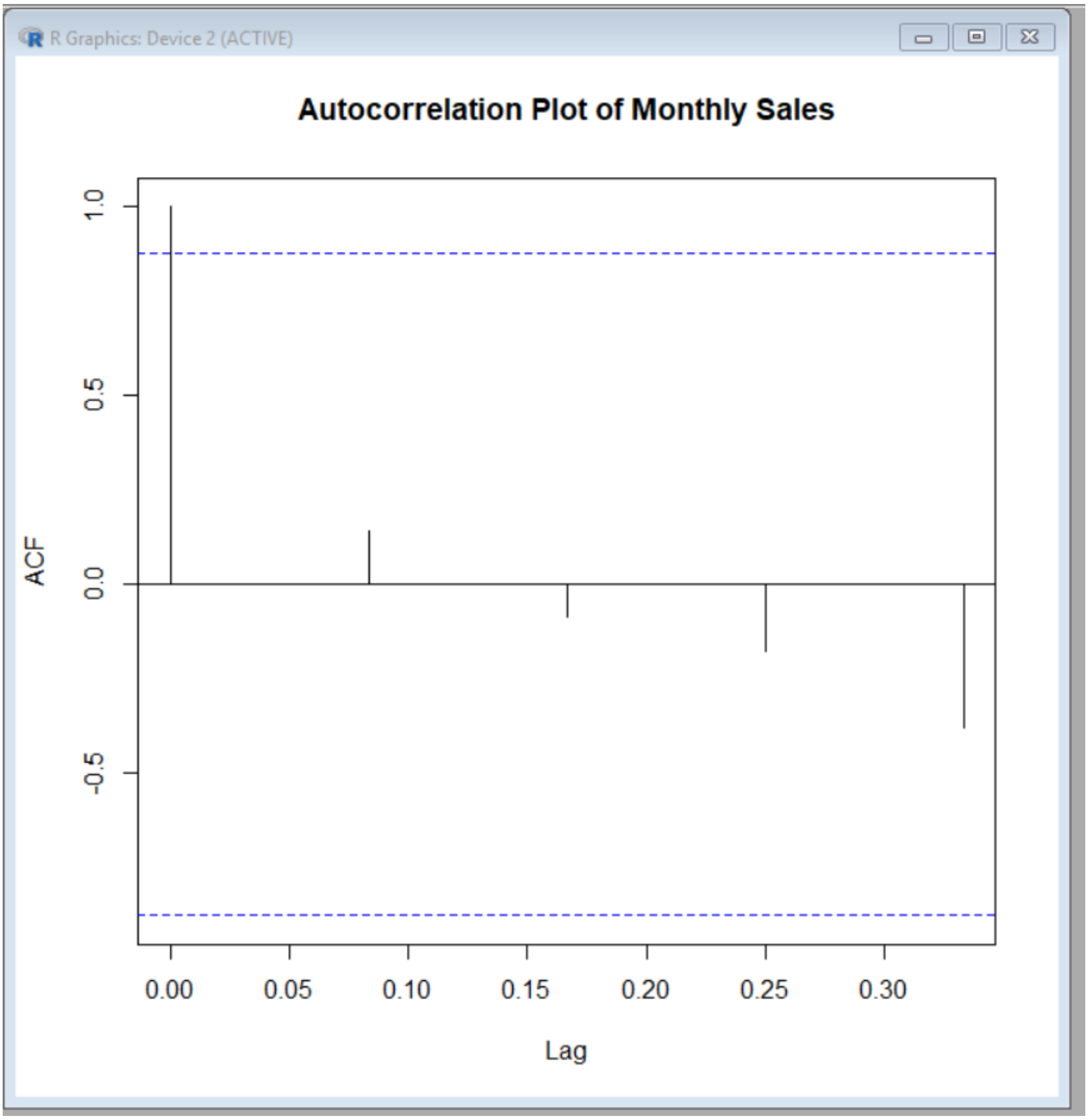


3. **In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.**



4. **In R, create an autocorrelation plot to identify seasonality in the time series data.**

```
sales <- c(15000, 18000, 22000, 20000, 23000)


sales_ts <- ts(sales,

            start = c(2026, 1),

            frequency = 12)


acf(sales_ts,

   main = "Autocorrelation Plot of Monthly Sales")
```



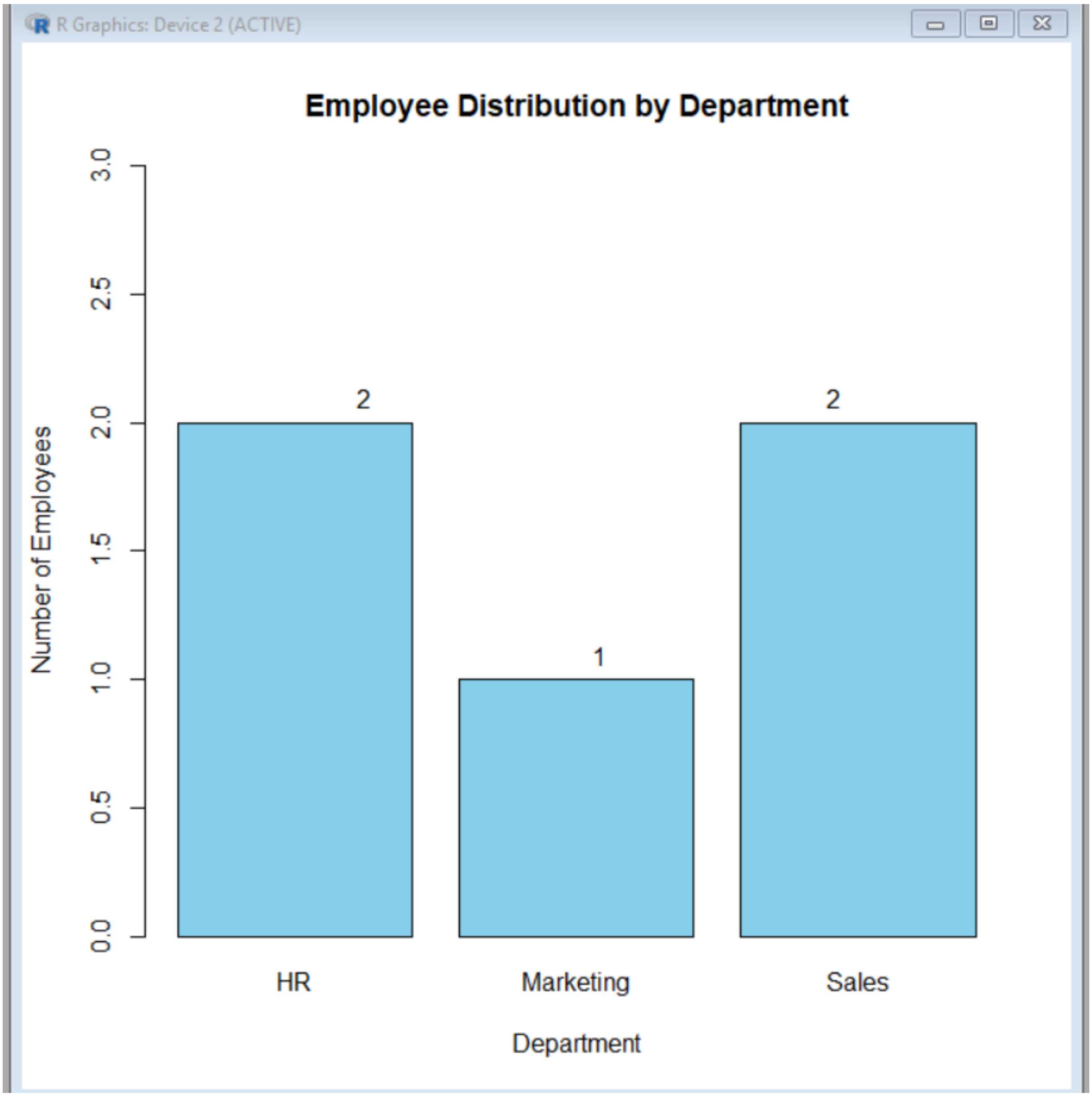
Set 23: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. **In R, create a bar chart to visualize the distribution of employees across different departments.Label the chart elements.**

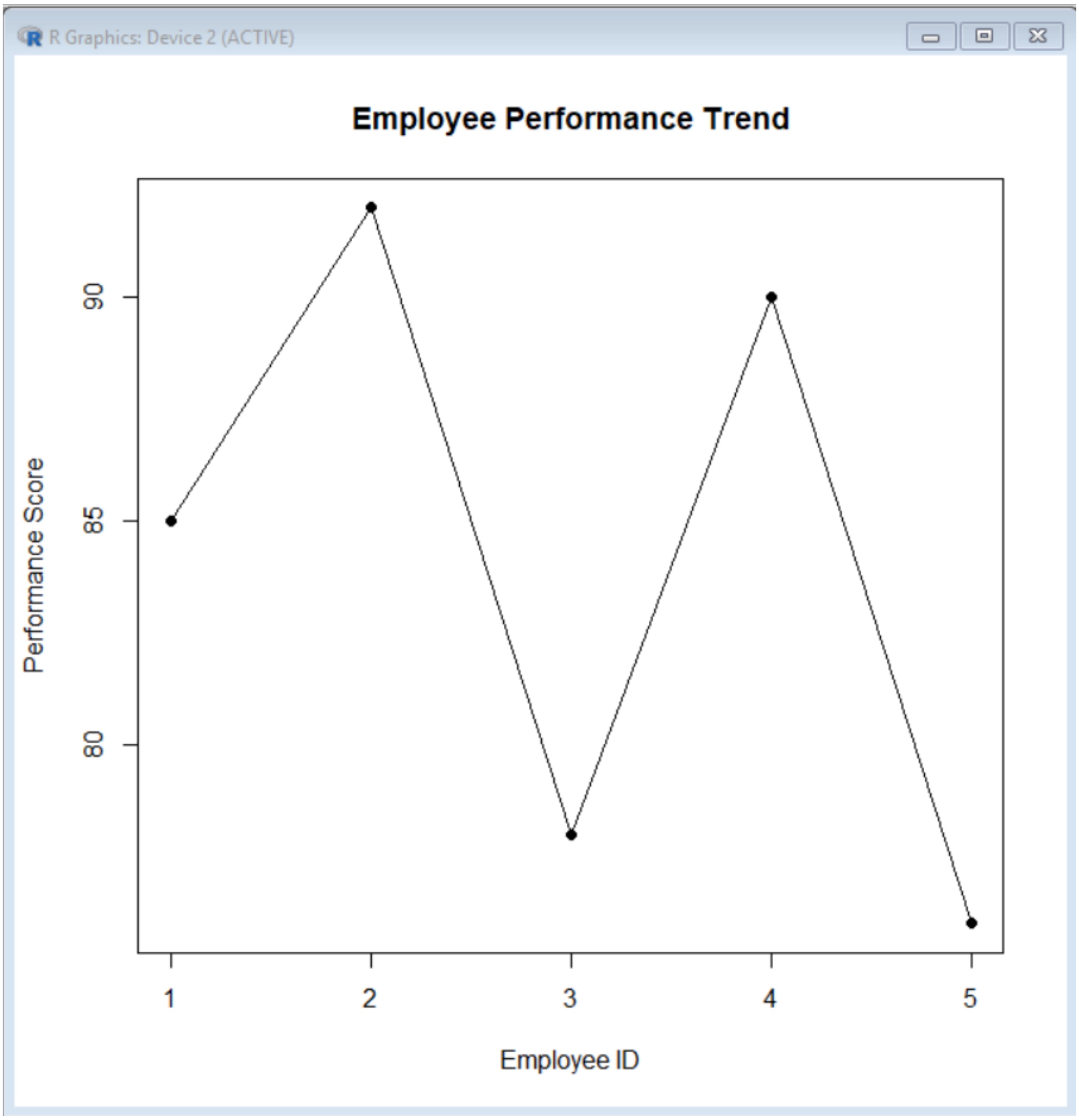
```
employee <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

dept_count <- table(employee$Department)

barplot(dept_count,
        main = "Employee Distribution by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        col = "skyblue",
        ylim = c(0, 3))

text(x = 1:length(dept_count),
     y = dept_count,
     labels = dept_count,
     pos = 3)
```



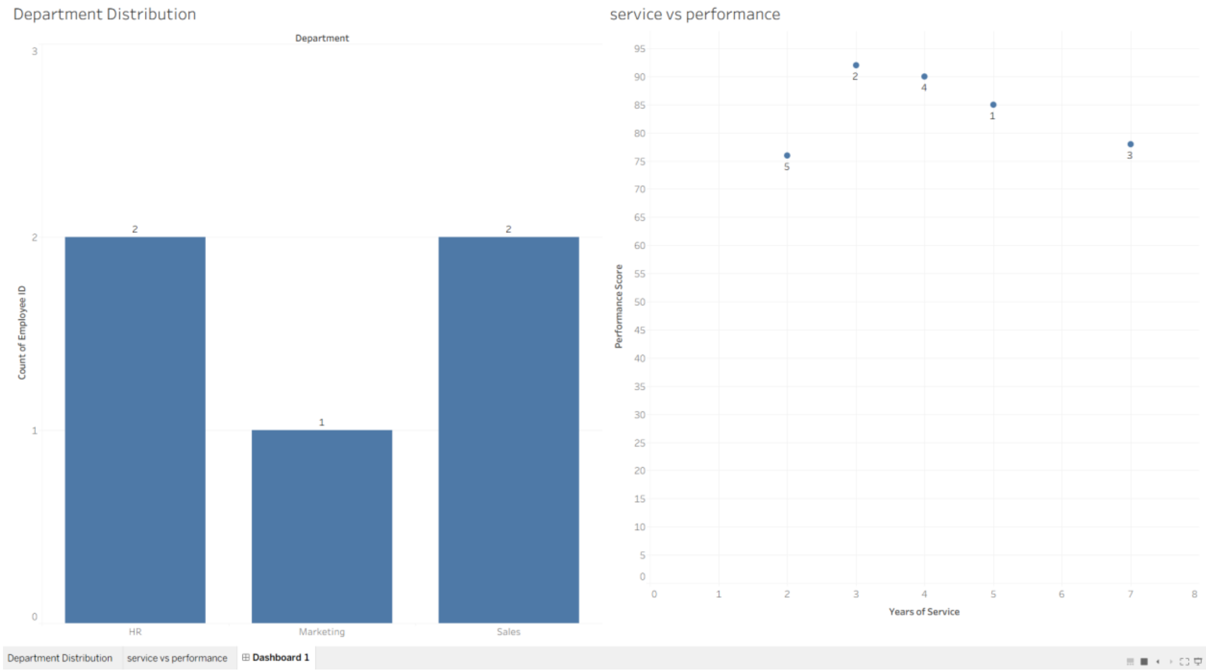
2. **In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.**

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

plot(employee$Employee_ID,
```

```
employee$Performance_Score,
type = "o",
main = "Employee Performance Trend",
xlab = "Employee ID",
ylab = "Performance Score",
pch = 19)
```



3. **In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.**



4. **In R, develop a table showing the performance data for each employee. Label the table elements.**

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)


print(employee)
```

```
Employee_ID Department Years_of_Service Performance_Score
          1      Sales                5                85
          2         HR                3                92
          3  Marketing                7                78
          4      Sales                4                90
          5         HR                2                76
```

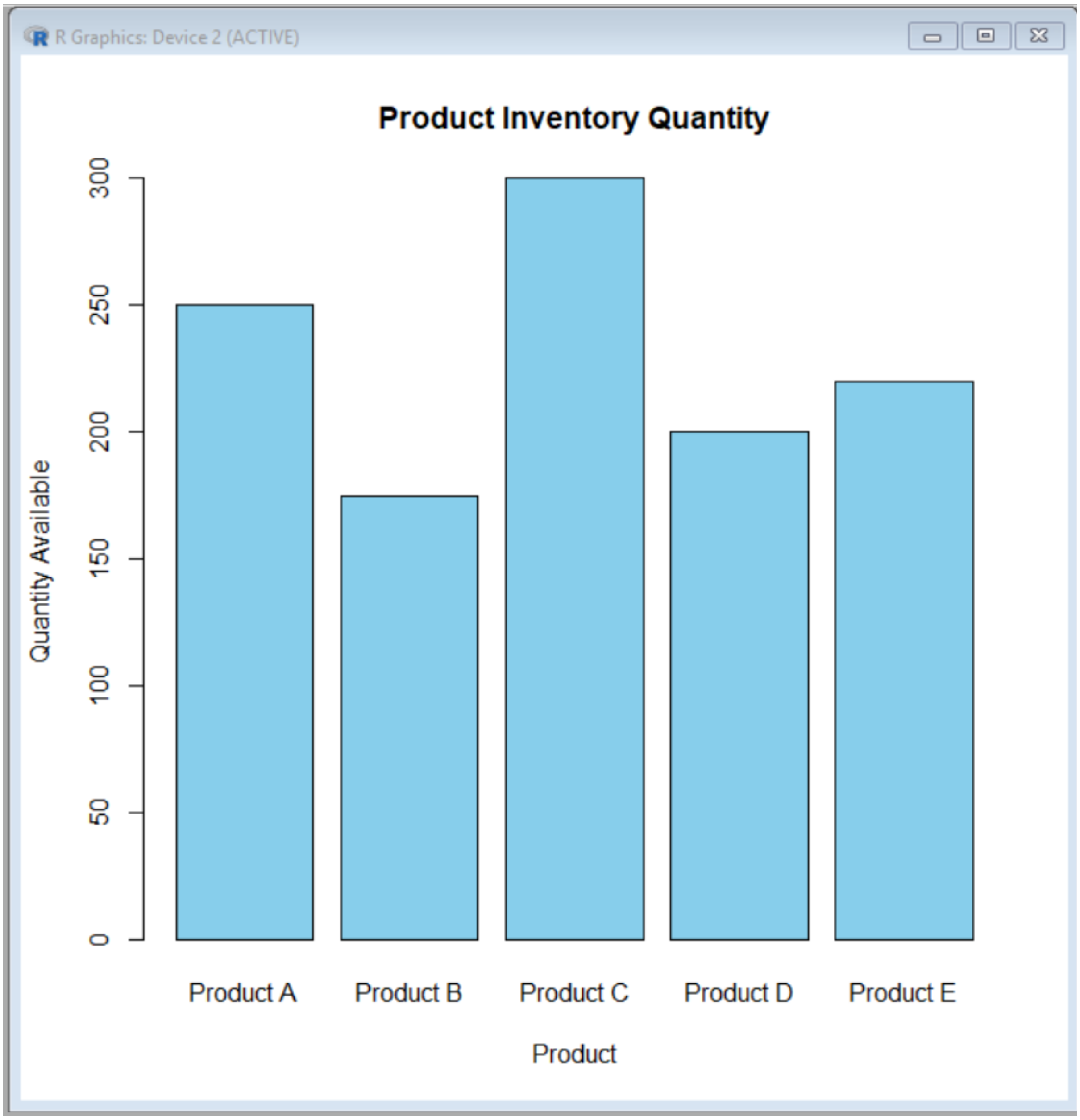
Set 24 Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1.In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

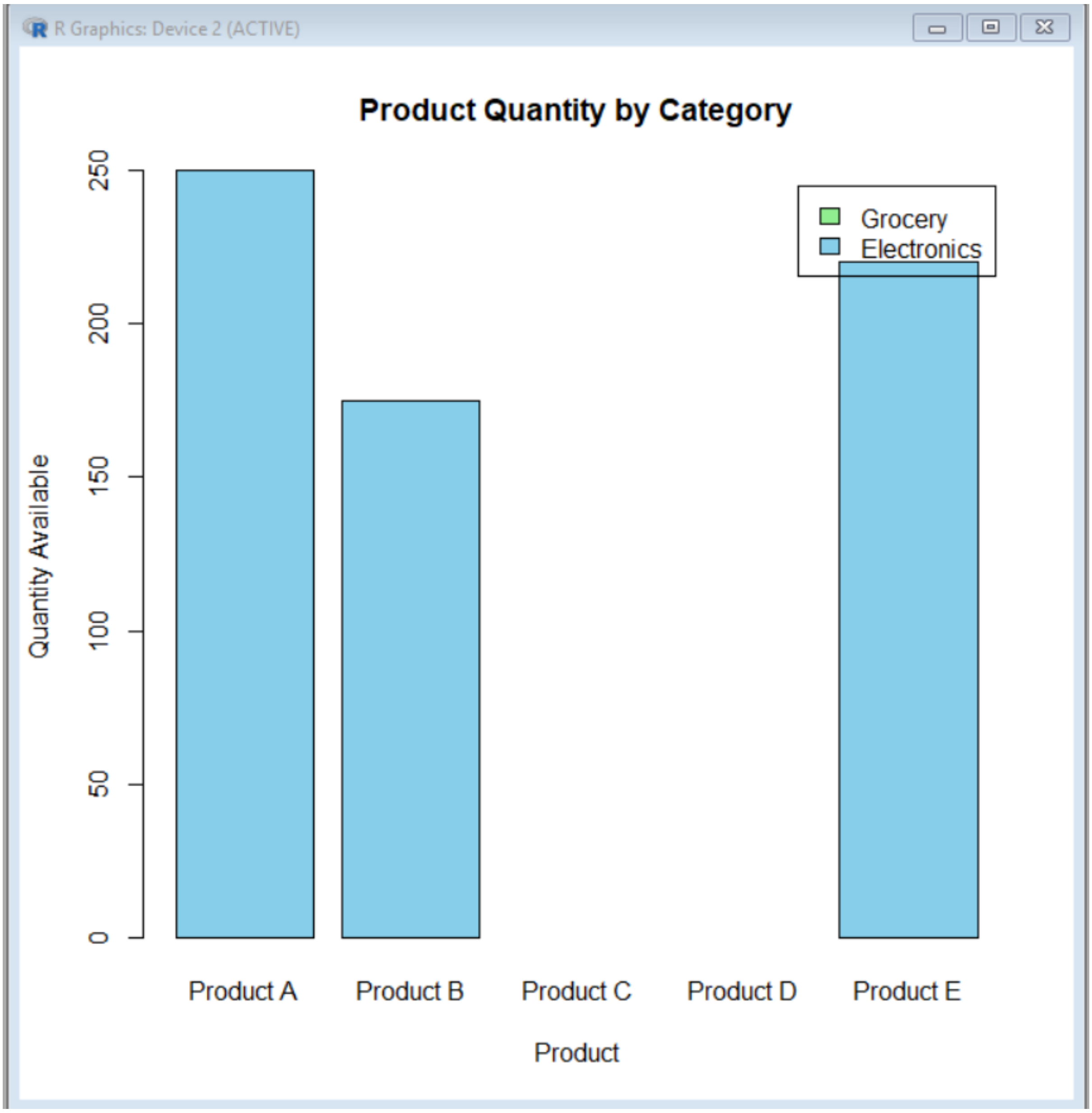
```
inventory <- data.frame(

  Product_ID = c(1, 2, 3, 4, 5),

  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),

  Quantity_Available = c(250, 175, 300, 200, 220)

)


barplot(inventory$Quantity_Available,

        names.arg = inventory$Product_Name,

        main = "Product Inventory Quantity",

        xlab = "Product",

        ylab = "Quantity Available",

        col = "skyblue")
```



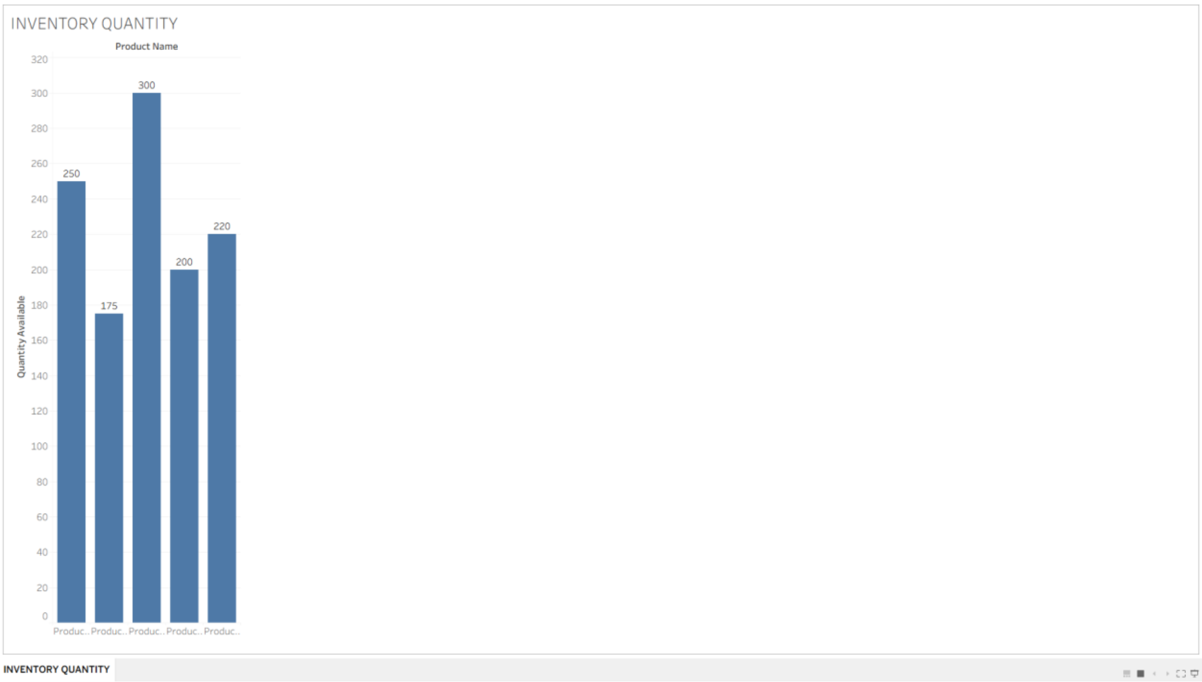
2.In R, generate a stacked bar chart to show the quantity of each product within different product categories.

```
inventory <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C",
                   "Product D", "Product E"),
  Category = c("Electronics", "Electronics", "Grocery",
               "Grocery", "Electronics"),
  Quantity = c(250, 175, 300, 200, 220)
)
```

```
barplot(tapply(inventory$Quantity,
          list(inventory$Category, inventory$Product_Name),
          sum),
     main = "Product Quantity by Category",
     xlab = "Product",
     ylab = "Quantity Available",
     col = c("skyblue", "lightgreen"),
     legend.text = TRUE)
```



3.In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.



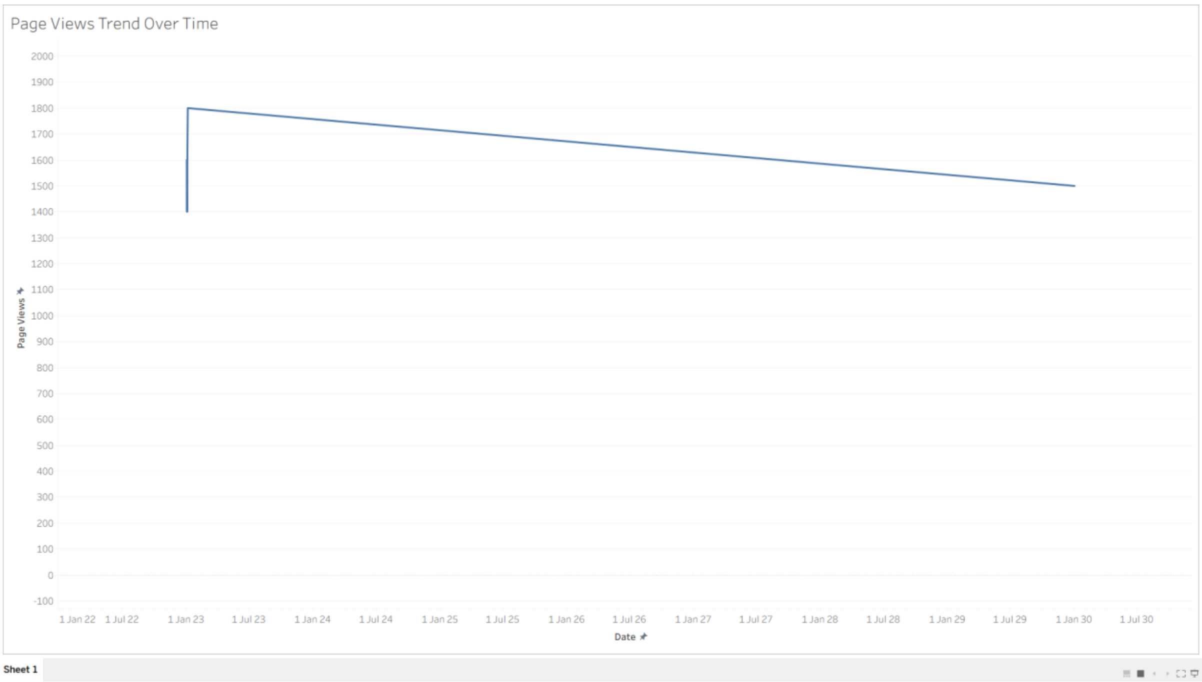
4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

A scatter plot between Product Price and Quantity Available cannot be created using the given dataset because Product Price is not provided. The dataset must contain a Product Price column to analyze the relationship between price and quantity available.

Set 25: Website Traffic Analysis

Dataset: Website Traffic

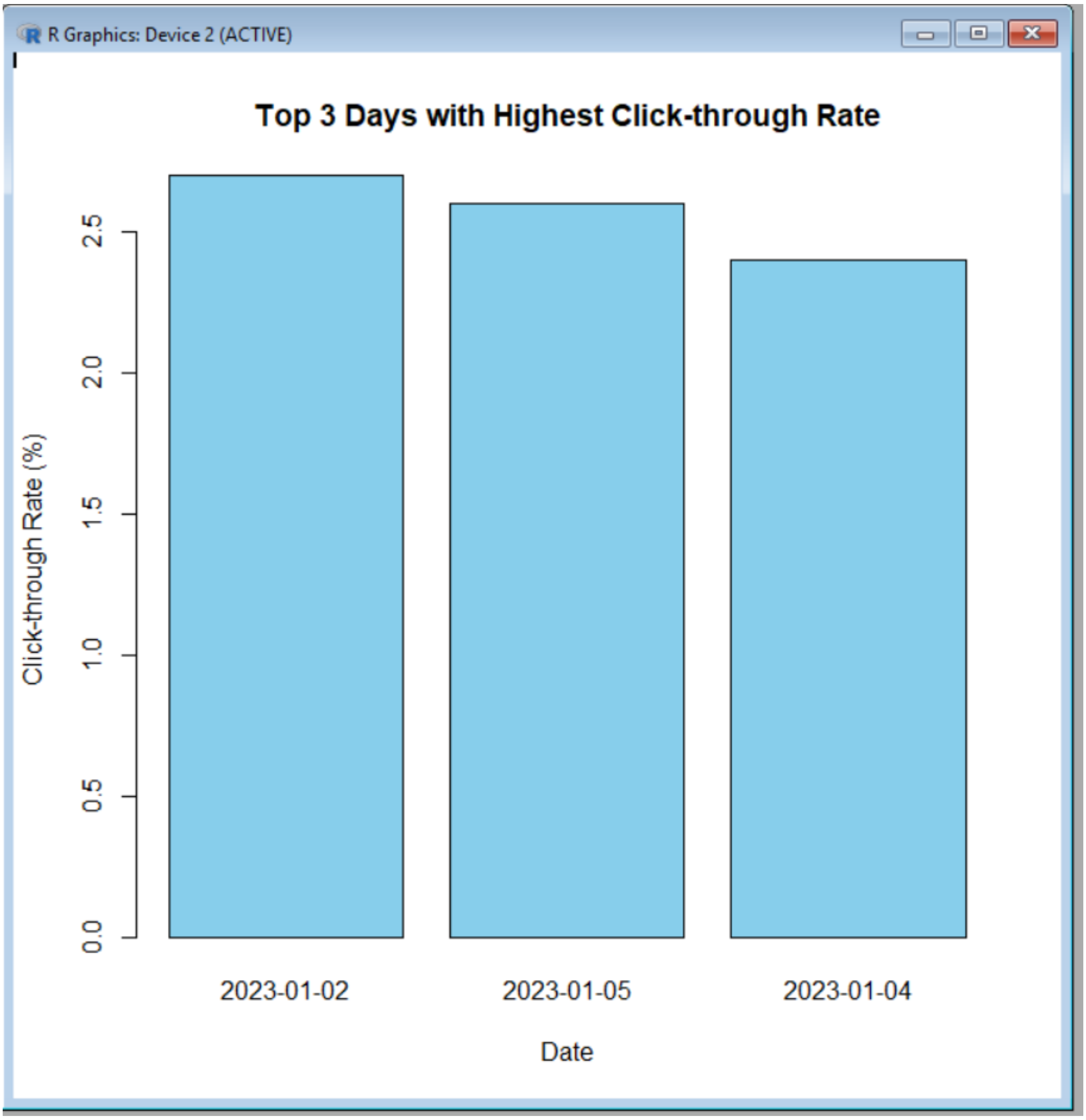
1.In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.



2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

```
# Create the data

date <- c("2023-01-01", "2023-01-02", "2023-01-03",

        "2023-01-04", "2023-01-05")


click_rate <- c(2.3, 2.7, 2.0, 2.4, 2.6)


# Create data frame

traffic <- data.frame(date, click_rate)


# Sort by click-through rate in descending order

traffic <- traffic[order(-traffic$click_rate), ]


# Select Top 3 days

top_n <- head(traffic, 3)


# Create bar chart
```

```
barplot(top_n$click_rate,
        names.arg = top_n$date,
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = "Top 3 Days with Highest Click-through Rate",
        col = "skyblue")
```

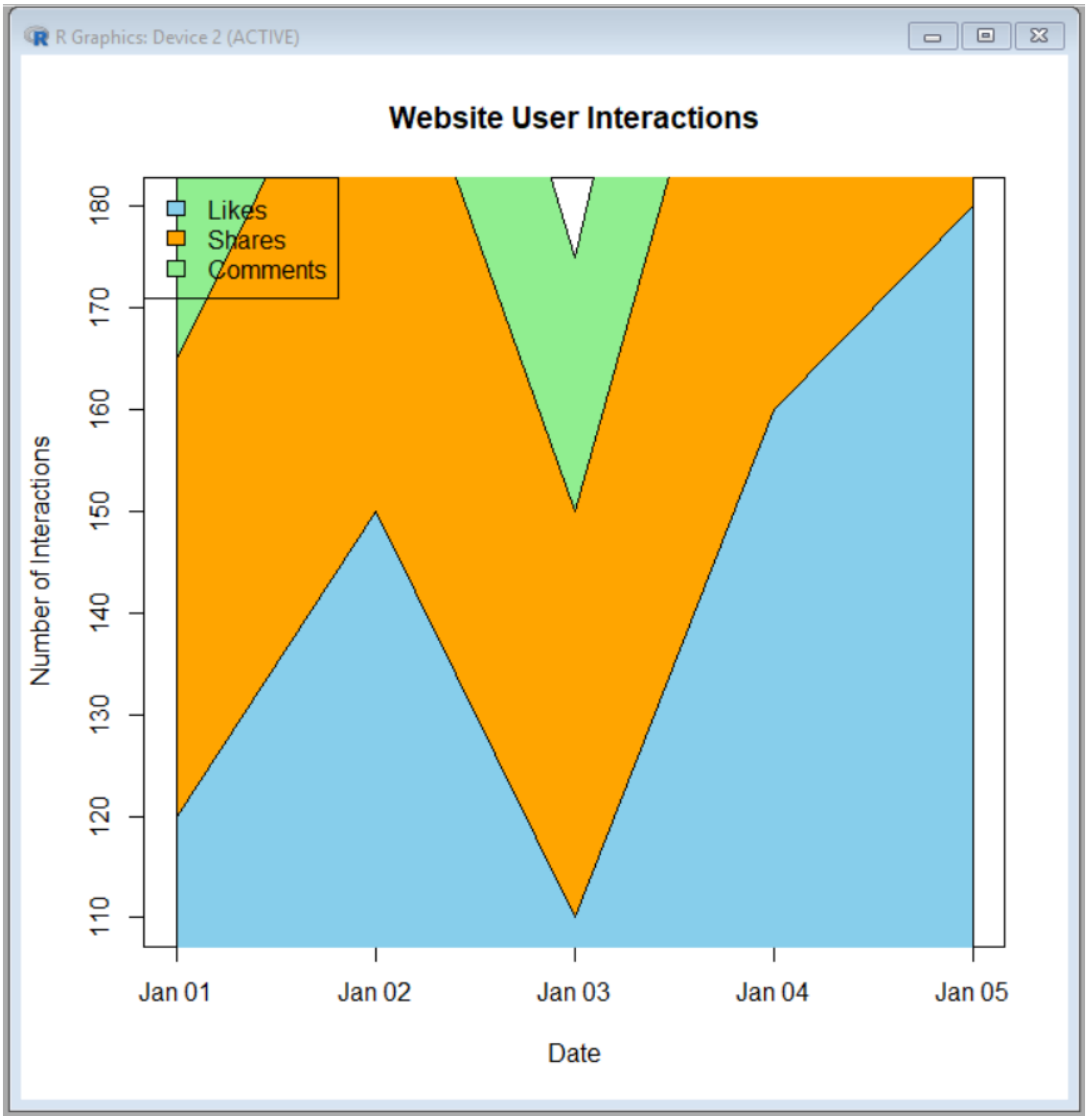


3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares,comments) on the website.

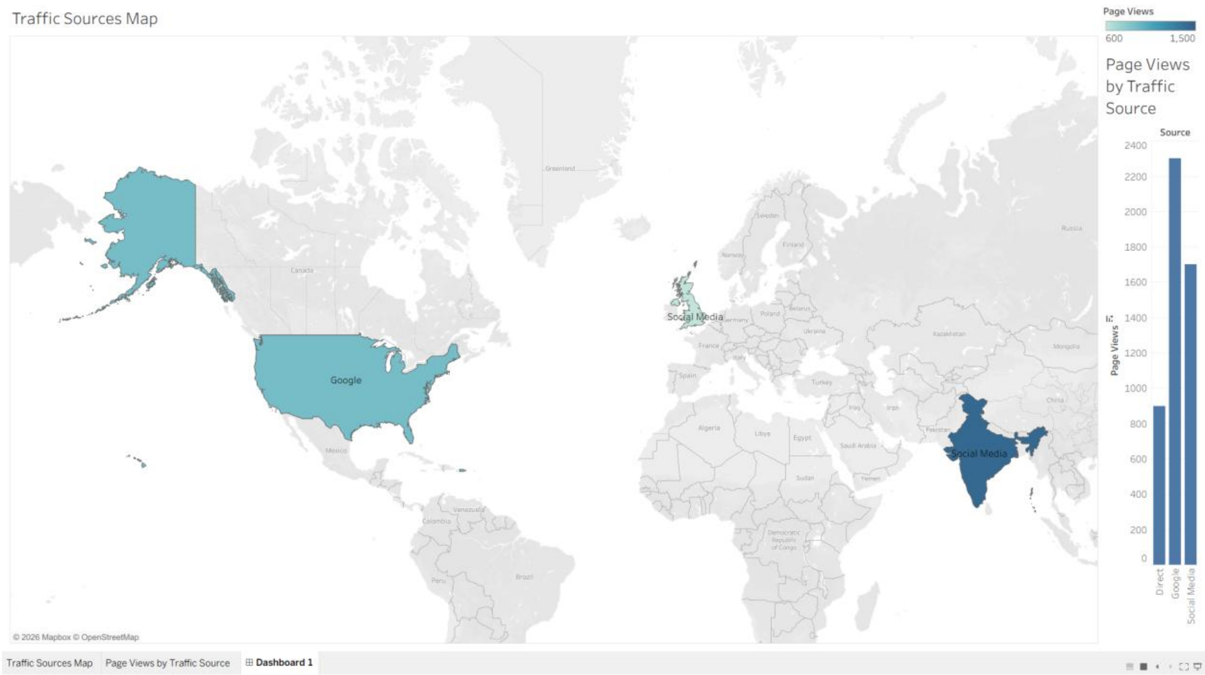
```
# Create the data
```

```r
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05"))

likes <- c(120, 150, 110, 160, 180)
shares <- c(45, 55, 40, 60, 70)
comments <- c(30, 35, 25, 40, 45)

# Create data frame
traffic <- data.frame(date, likes, shares, comments)

# Create stacked area chart
plot(traffic$date, traffic$likes,
     type = "n",
     xlab = "Date",
     ylab = "Number of Interactions",
     main = "Website User Interactions")

polygon(c(traffic$date, rev(traffic$date)),
        c(traffic$likes, rep(0, nrow(traffic))),
        col = "skyblue")

polygon(c(traffic$date, rev(traffic$date)),
        c(traffic$likes + traffic$shares,
          rev(traffic$likes)),
        col = "orange")

polygon(c(traffic$date, rev(traffic$date)),
        c(traffic$likes + traffic$shares + traffic$comments,
```

```
    rev(traffic$likes + traffic$shares)),
  col = "lightgreen")


legend("topleft",
   legend = c("Likes", "Shares", "Comments"),
   fill = c("skyblue", "orange", "lightgreen"))
```



4.In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

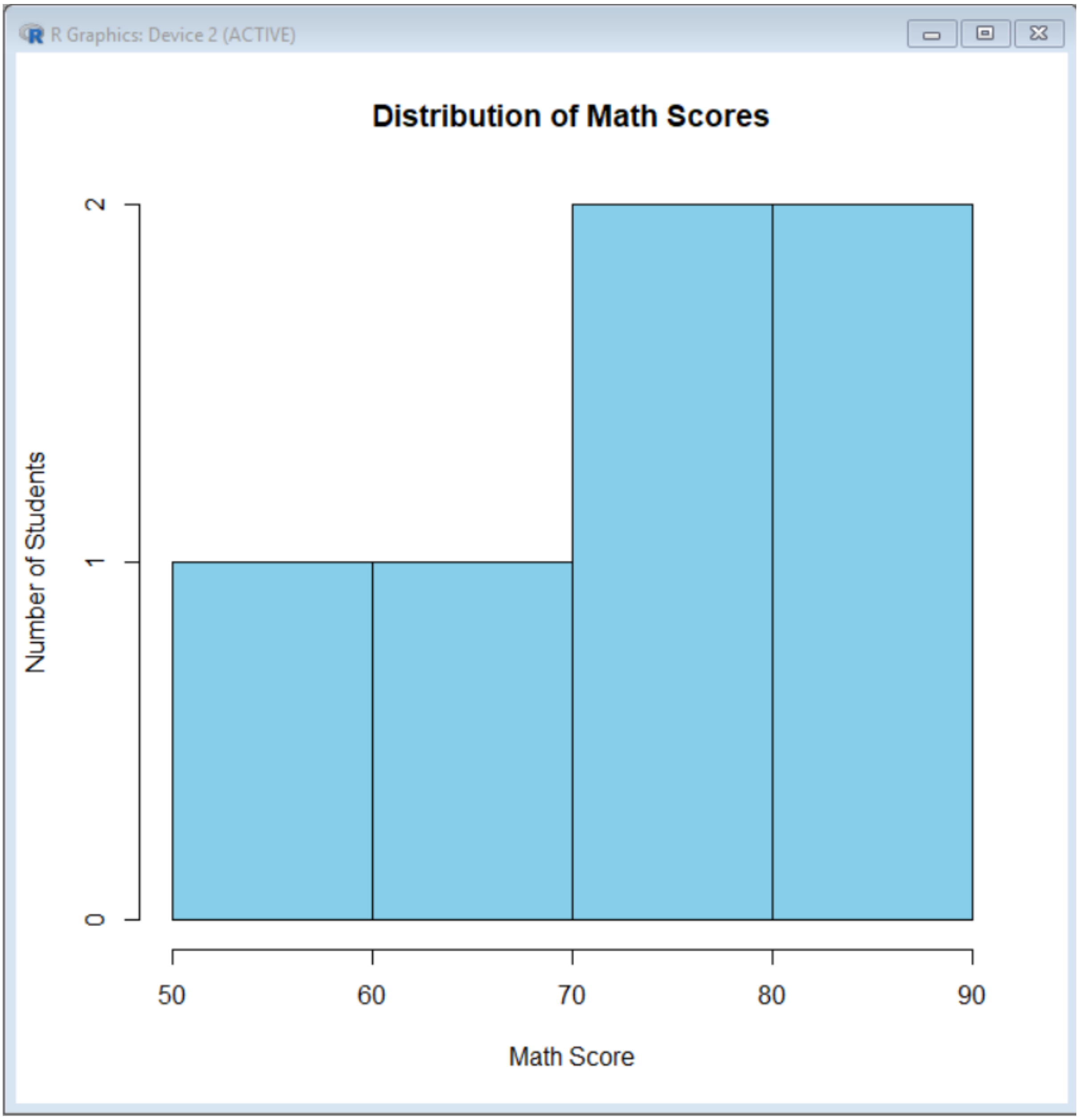


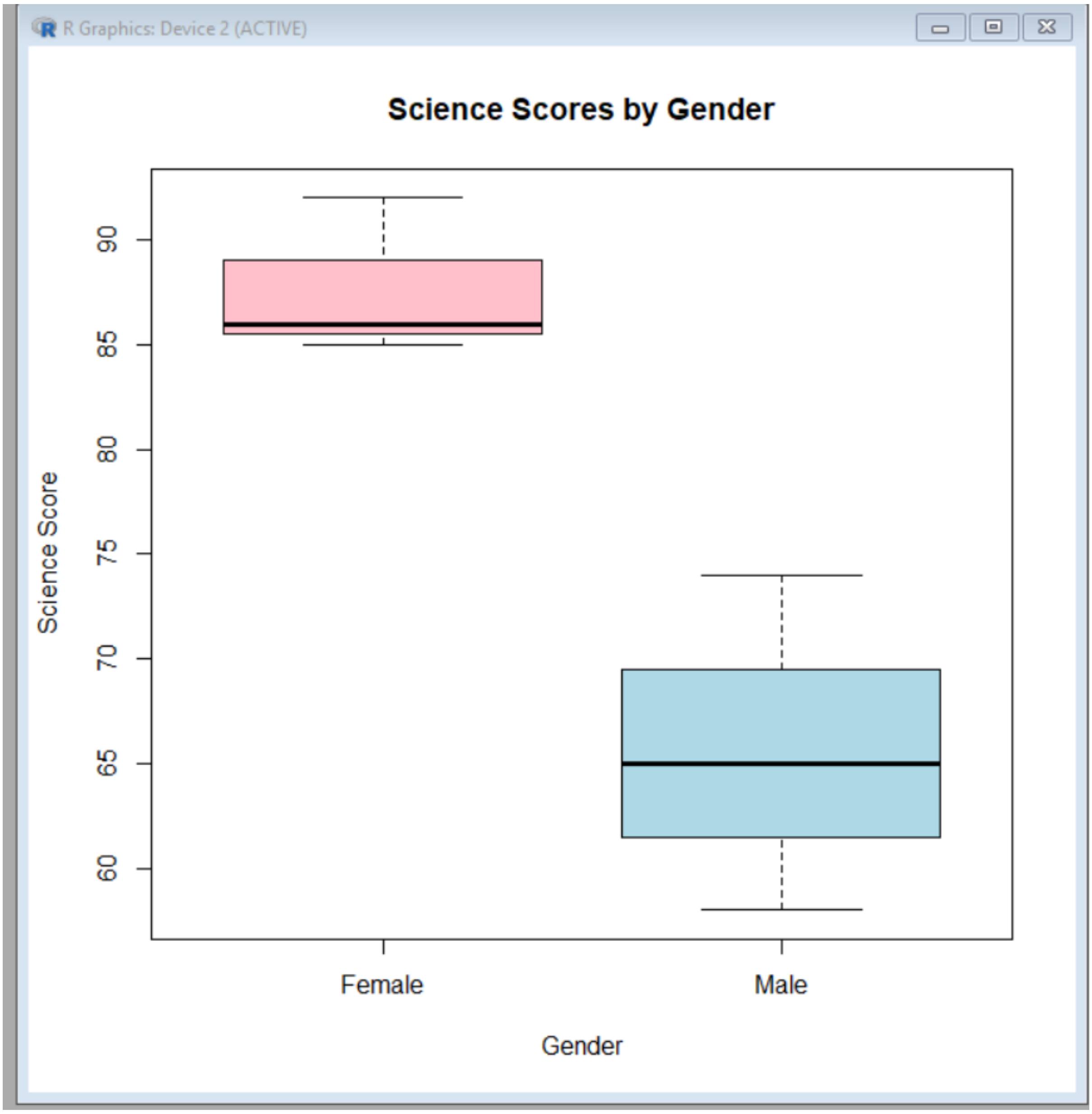
SET 26

Dataset: Student_Mini_Data.csv

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

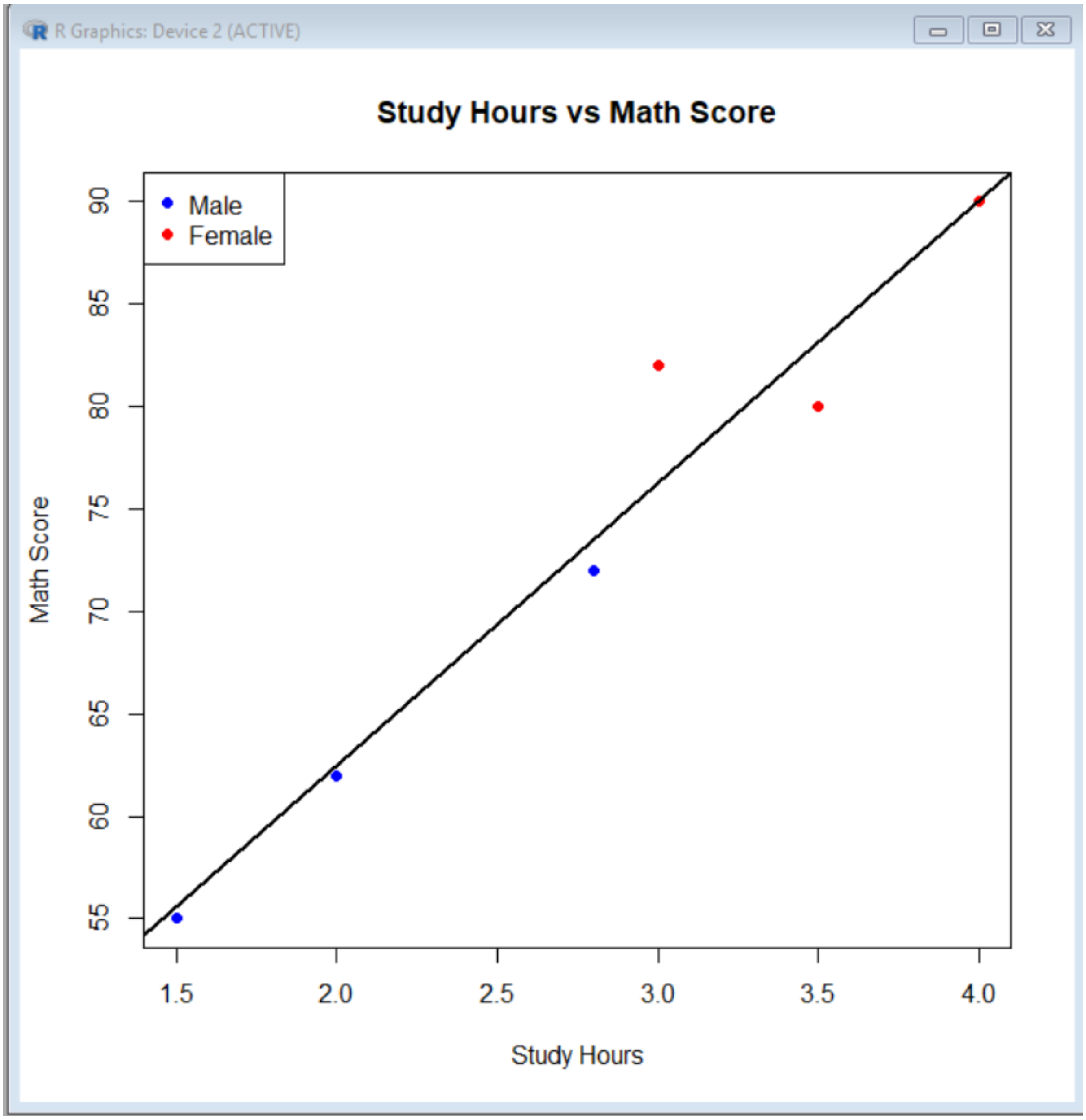

R Graphics: Device 2 (ACTIVE)
Distribution of Math Scores
Number of Students
0
1
2
50
60
70
80
90
Math Score



2. Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

```
# Create scatter plot
plot(students$Study_Hours, students$Math_Score,
    col = ifelse(students$Gender == "Male", "blue", "red"),
    pch = 19,
    xlab = "Study Hours",
    ylab = "Math Score",
```

```
     main = "Study Hours vs Math Score")

# Add regression line
abline(lm(Math_Score ~ Study_Hours, data = students),
       col = "black",
       lwd = 2)

# Add legend
legend("topleft",
       legend = c("Male", "Female"),
       col = c("blue", "red"),
       pch = 19)
```



3. Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

Convert Exam_Date into Date format

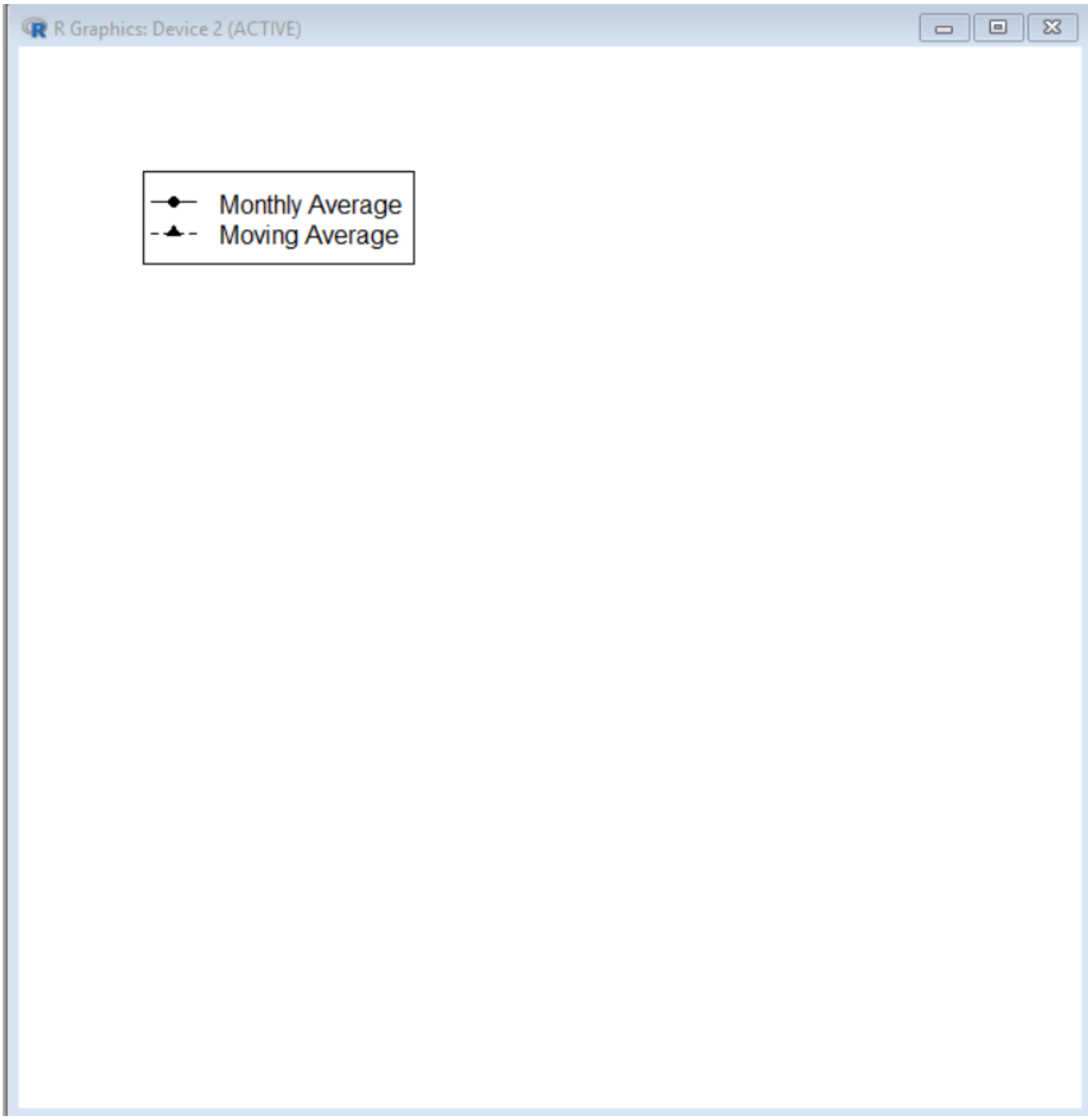
students$Exam_Date <- as.Date(students$Exam_Date)

Extract month

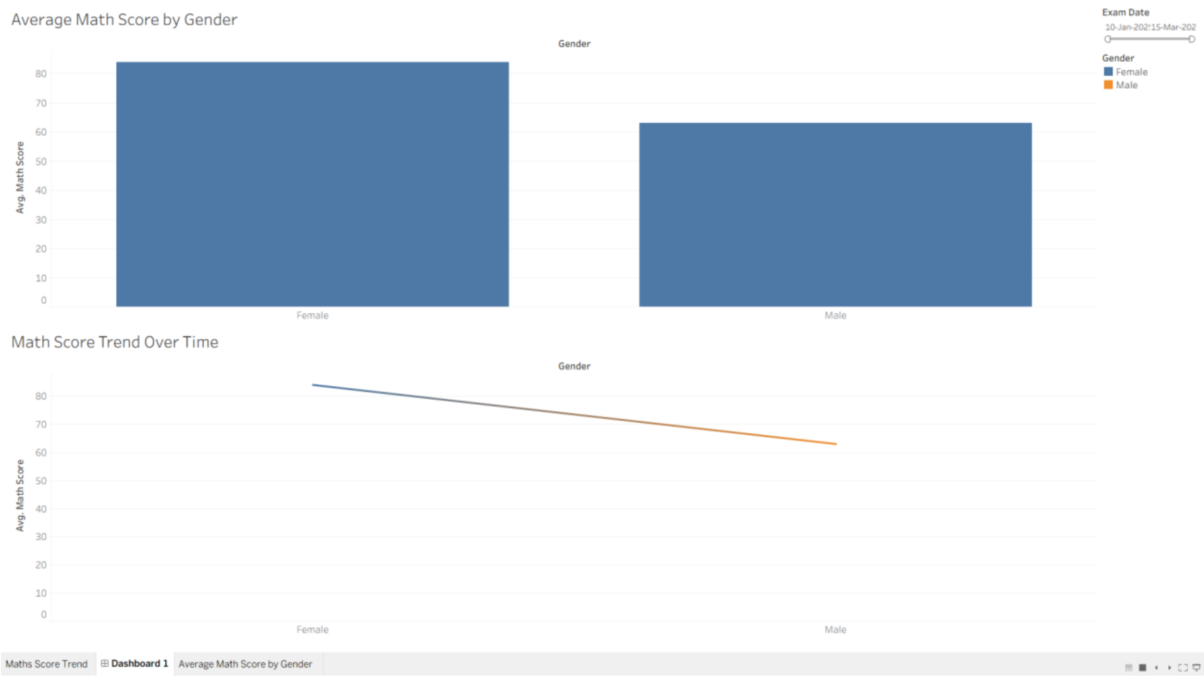
students$Month <- format(students$Exam_Date, "%Y-%m")

```r
# Calculate monthly average Math Score
monthly_avg <- aggregate(Math_Score ~ Month,
                         data = students,
                         FUN = mean)

print(monthly_avg)

# Calculate 2-month moving average
monthly_avg$Moving_Average <- c(
  NA,
  (monthly_avg$Math_Score[1:2] + monthly_avg$Math_Score[2:3]) / 2
)

print(monthly_avg)

# Plot monthly average Math Score
plot(monthly_avg$Month,
     monthly_avg$Math_Score,
     type = "o",
     pch = 19,
     xlab = "Month",
     ylab = "Average Math Score",
     main = "Monthly Average Math Score")

# Add moving average line
lines(monthly_avg$Month,
      monthly_avg$Moving_Average,
      type = "o",
```

```
      pch = 17,
      lty = 2)

# Add legend
legend("topleft",
       legend = c("Monthly Average", "Moving Average"),
       lty = c(1, 2),
       pch = c(19, 17))
```



4. Import Student_Mini_Data.csv into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.



SET 27.

Patient Health Risk Analysis

Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol.Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

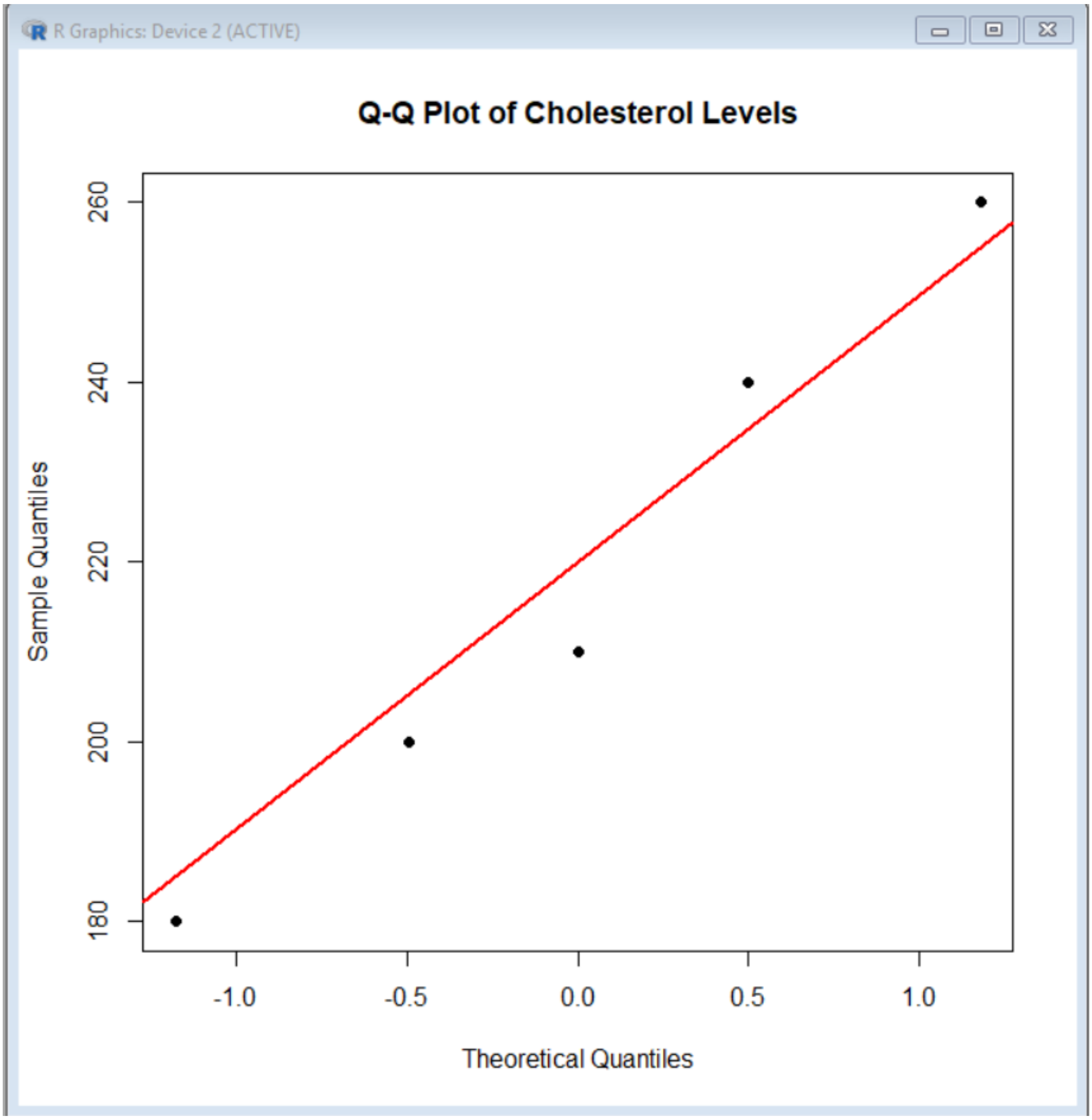
```
# Create the dataset
health <- data.frame(
  Patient_ID = c("P1", "P2", "P3", "P4", "P5"),
  Age = c(25, 40, 55, 35, 60),
```

```
  BMI = c(22, 28, 30, 26, 32),
  BP = c(120, 135, 145, 130, 150),
  Cholesterol = c(180, 210, 240, 200, 260)
)

# Create scatterplot matrix
pairs(health[, c("Age", "BMI", "BP", "Cholesterol")],
     main = "Scatterplot Matrix of Patient Health Indicators",
     pch = 19)
```

2.Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour.

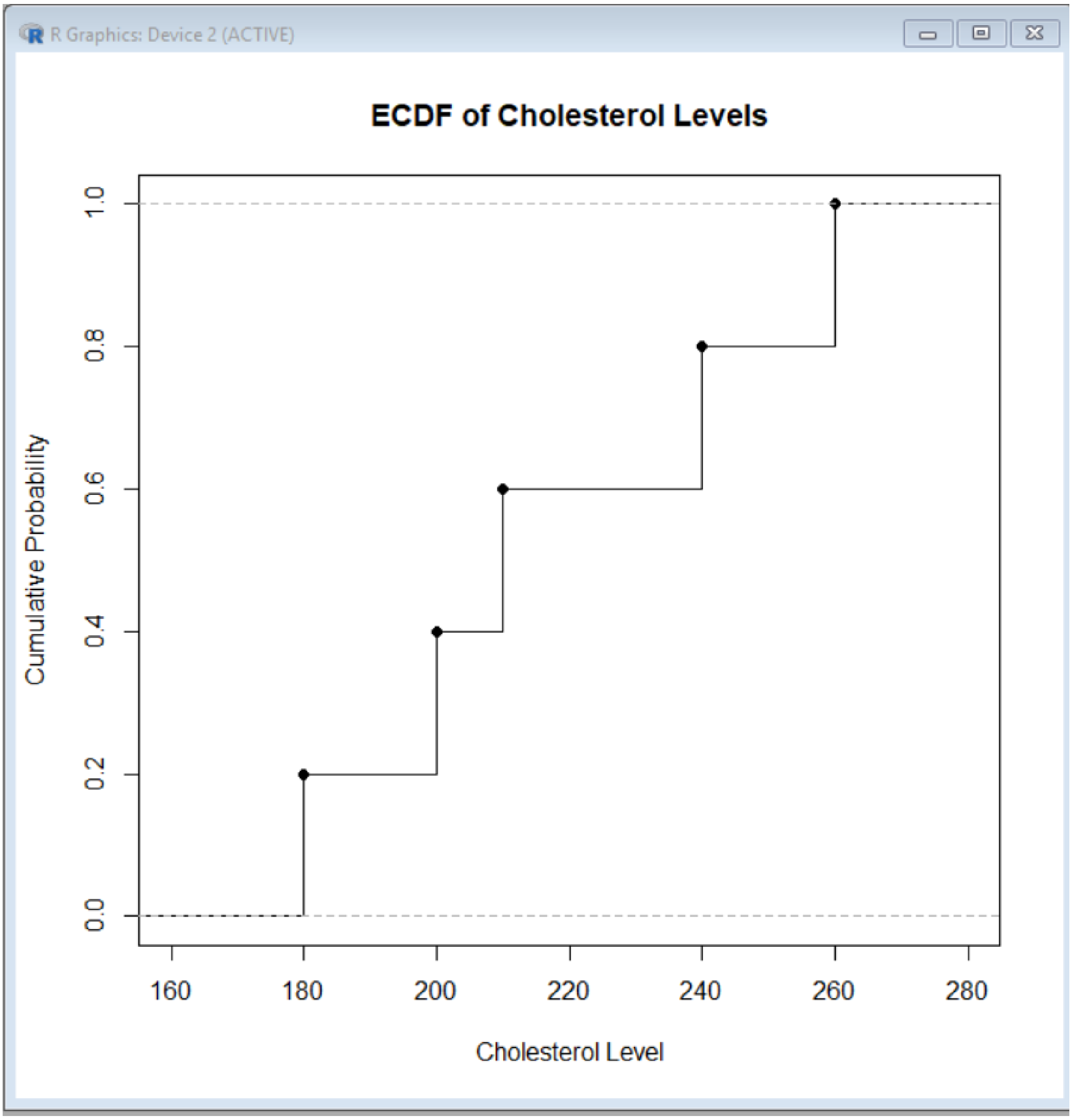
Comment on normality assumption and skewness.

```
# Q-Q plot
qqnorm(health$Cholesterol,
    main = "Q-Q Plot of Cholesterol Levels",
    xlab = "Theoretical Quantiles",
    ylab = "Sample Quantiles",
    pch = 19)

# Add reference line
qqline(health$Cholesterol,
    col = "red",
    lwd = 2)
```



2. ECDF for Cholesterol

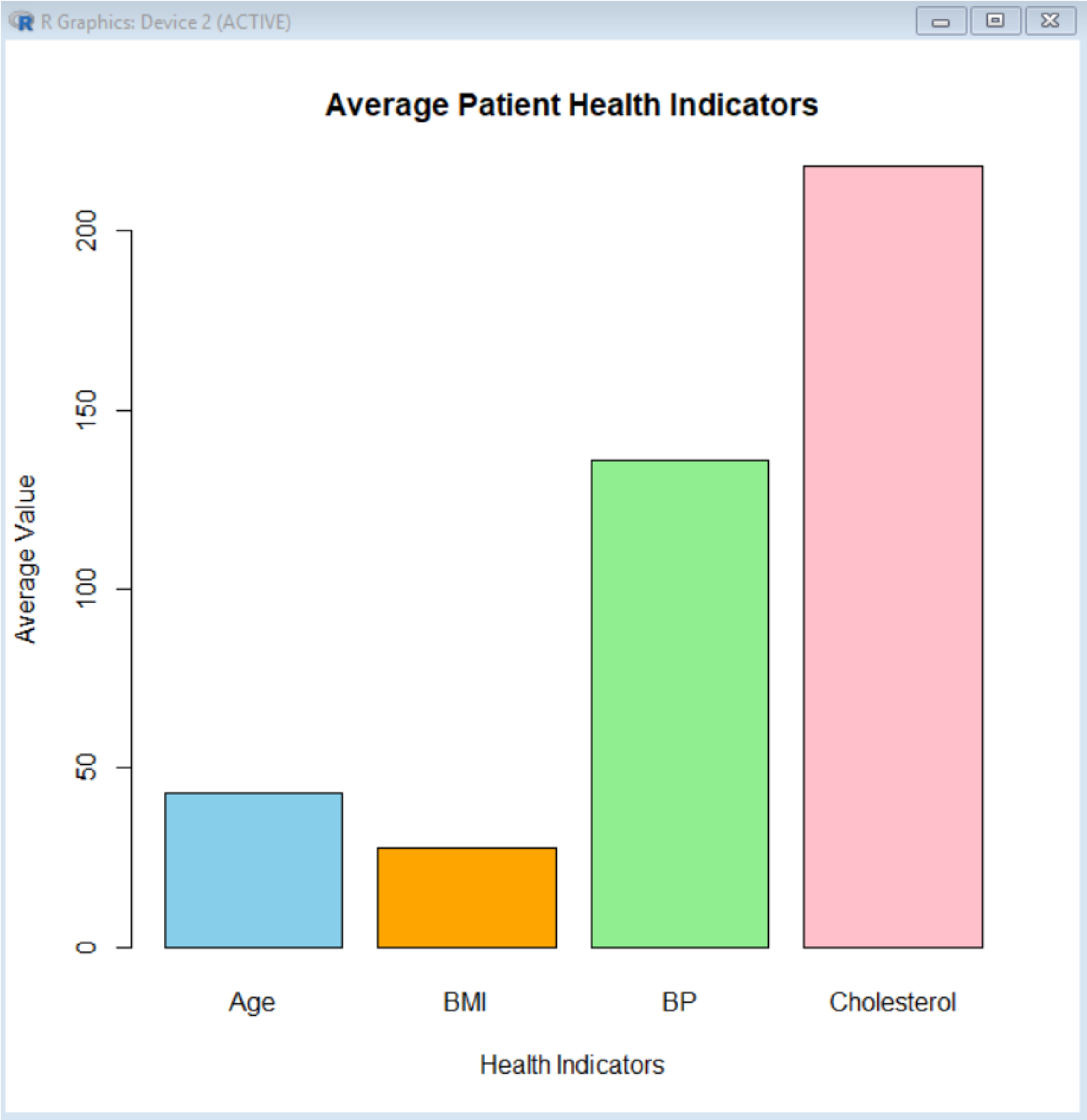
```
# ECDF plot

plot(ecdf(health$Cholesterol),

    main = "ECDF of Cholesterol Levels",

    xlab = "Cholesterol Level",

    ylab = "Cumulative Probability",

    verticals = TRUE,

    do.points = TRUE)
```



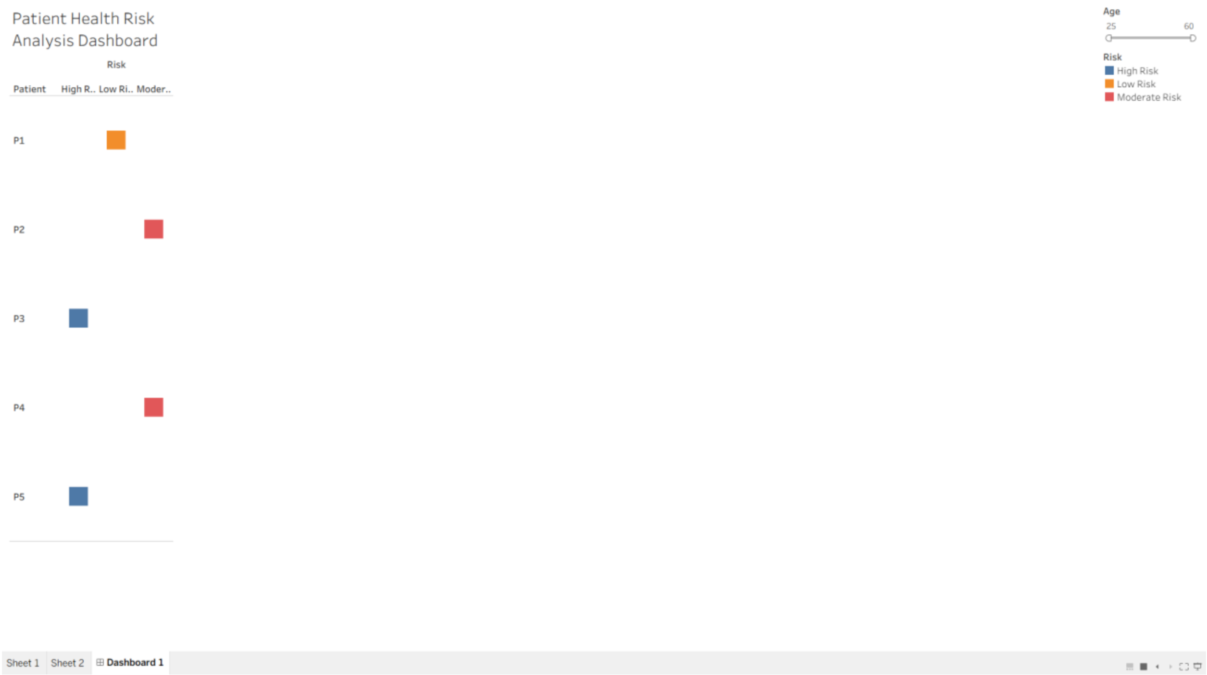
3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI,BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

```
# Calculate average values

averages <- c(

  Age = mean(health$Age),

  BMI = mean(health$BMI),

  BP = mean(health$BP),
```

```
  Cholesterol = mean(health$Cholesterol)
)

# Display averages
print(averages)

# Create bar chart
barplot(averages,
        main = "Average Patient Health Indicators",
        xlab = "Health Indicators",
        ylab = "Average Value",
        col = c("skyblue", "orange", "lightgreen", "pink"),
        border = "black")
```



4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.



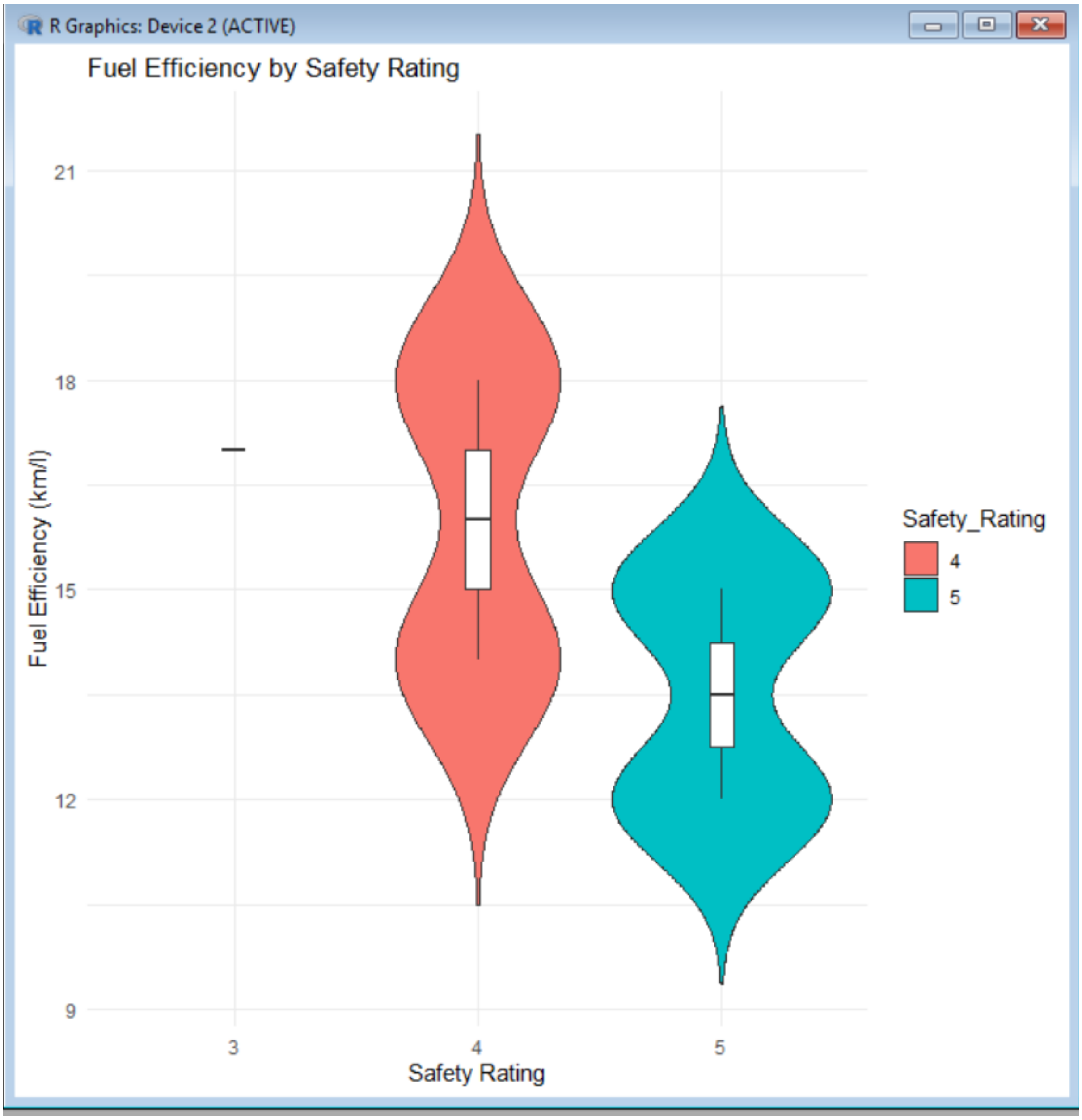
SET 28. Vehicle Performance Analysis

Vehicle_ID	Engine_Size (L)	Horsepower	Fuel_Efficiency (km/l)	Top_Speed (km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

```
# Install ggplot2 if needed
install.packages("ggplot2")

# Load package
library(ggplot2)

# Create dataset
vehicle <- data.frame(
```

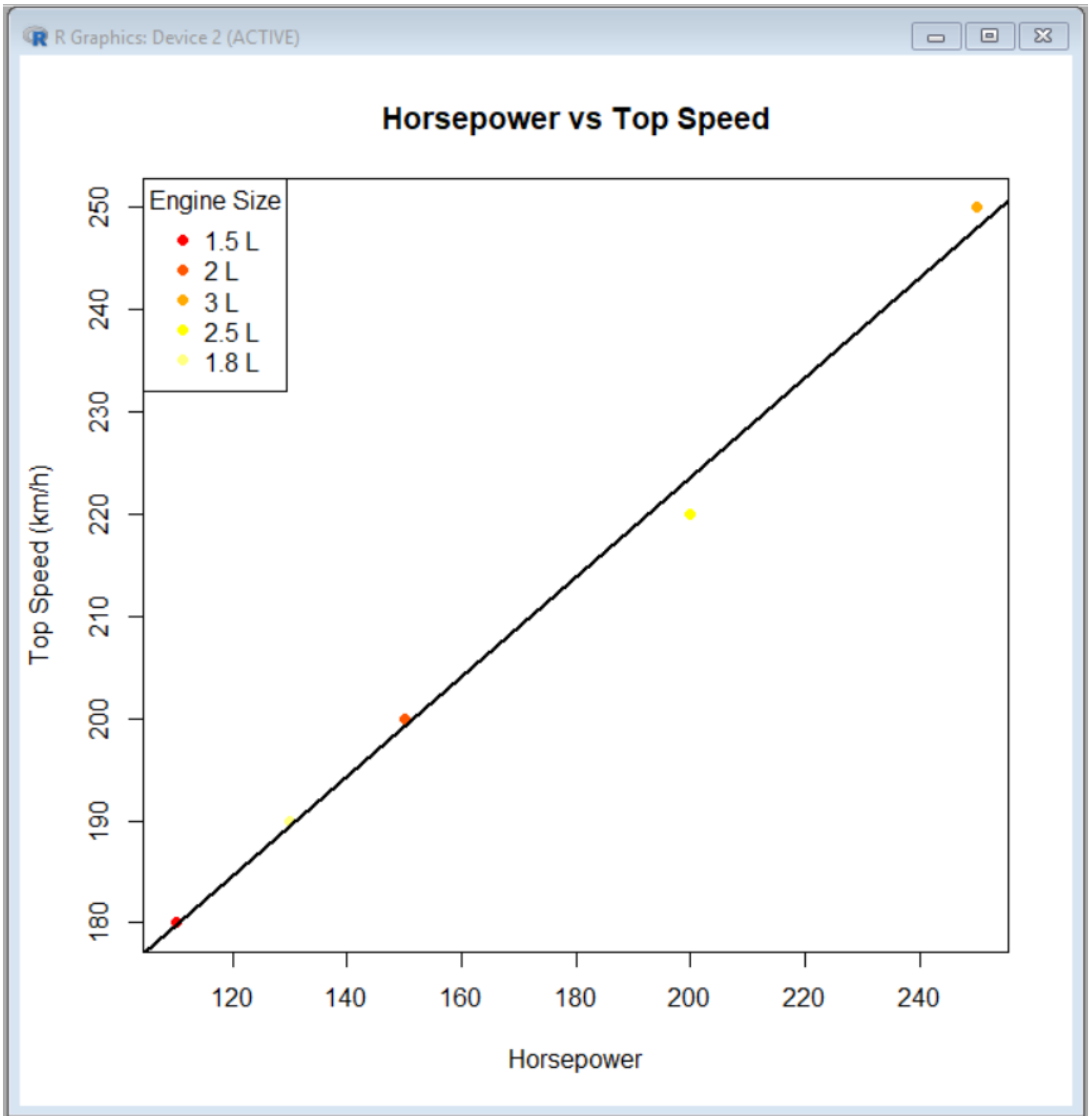
```
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed = c(180, 200, 250, 220, 190),
  Safety_Rating = c(4, 5, 5, 4, 3)
)

# Convert Safety_Rating to categorical variable
vehicle$Safety_Rating <- as.factor(vehicle$Safety_Rating)

# Create violin plot
ggplot(vehicle, aes(x = Safety_Rating,
                    y = Fuel_Efficiency,
                    fill = Safety_Rating)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1, fill = "white") +
  labs(
    title = "Fuel Efficiency by Safety Rating",
    x = "Safety Rating",
    y = "Fuel Efficiency (km/l)"
  ) +
  theme_minimal()
```



2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

```
# Scatter plot
plot(vehicle$Horsepower,
    vehicle$Top_Speed,
    col = heat.colors(length(vehicle$Engine_Size)),
    pch = 19,
    xlab = "Horsepower",
```

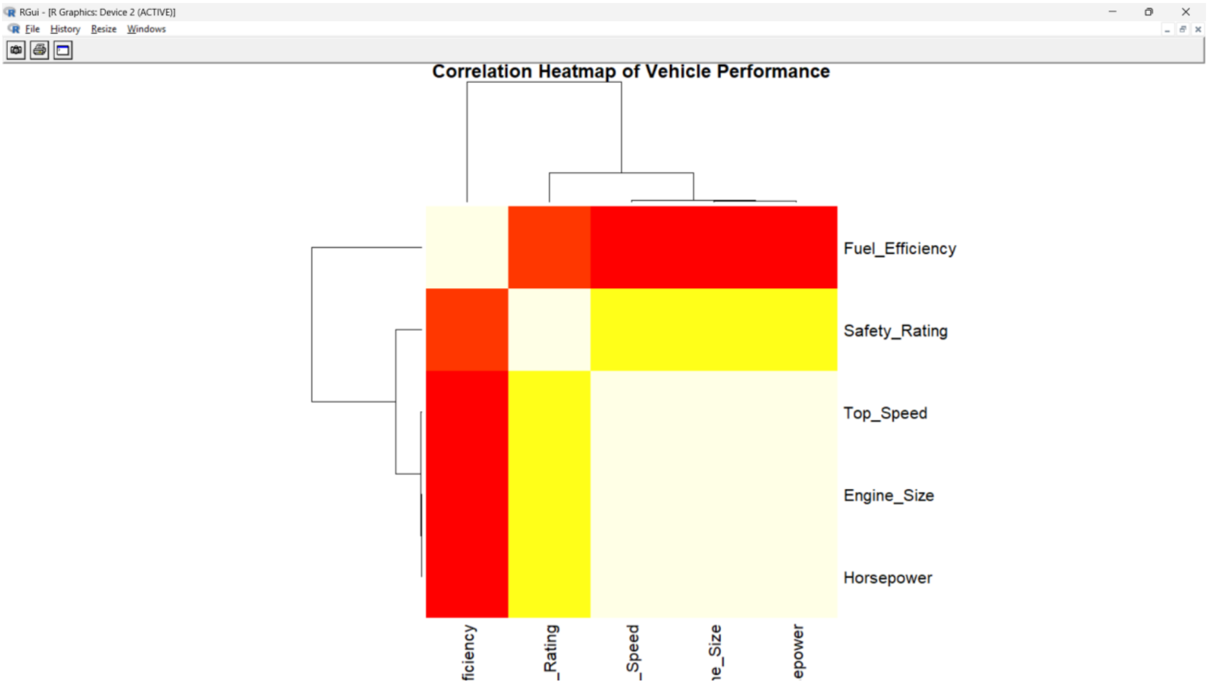
```r
    ylab = "Top Speed (km/h)",
    main = "Horsepower vs Top Speed")

# Add regression line
abline(lm(Top_Speed ~ Horsepower, data = vehicle),
       col = "black",
       lwd = 2)

# Add legend for Engine Size
legend("topleft",
       legend = paste(vehicle$Engine_Size, "L"),
       col = heat.colors(length(vehicle$Engine_Size)),
       pch = 19,
       title = "Engine Size")
```



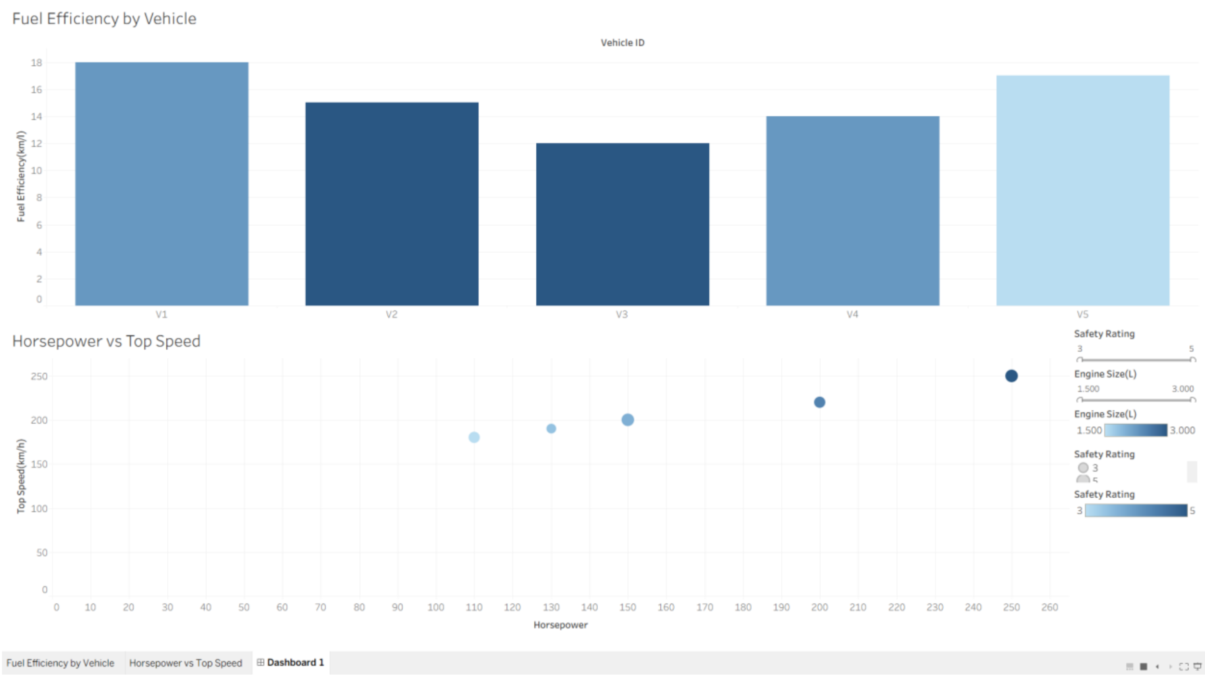
3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

```
# Select numerical variables
num_data <- vehicle[, c("Engine_Size",
                        "Horsepower",
                        "Fuel_Efficiency",
                        "Top_Speed",
                        "Safety_Rating")]
```

```
# Calculate correlation matrix
cor_matrix <- cor(num_data)

# Display correlation values
print(cor_matrix)

# Create correlation heatmap
heatmap(cor_matrix,
    main = "Correlation Heatmap of Vehicle Performance",
    col = heat.colors(20),
    symm = TRUE)
```



4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.



SET 29 .Student Academic Performance

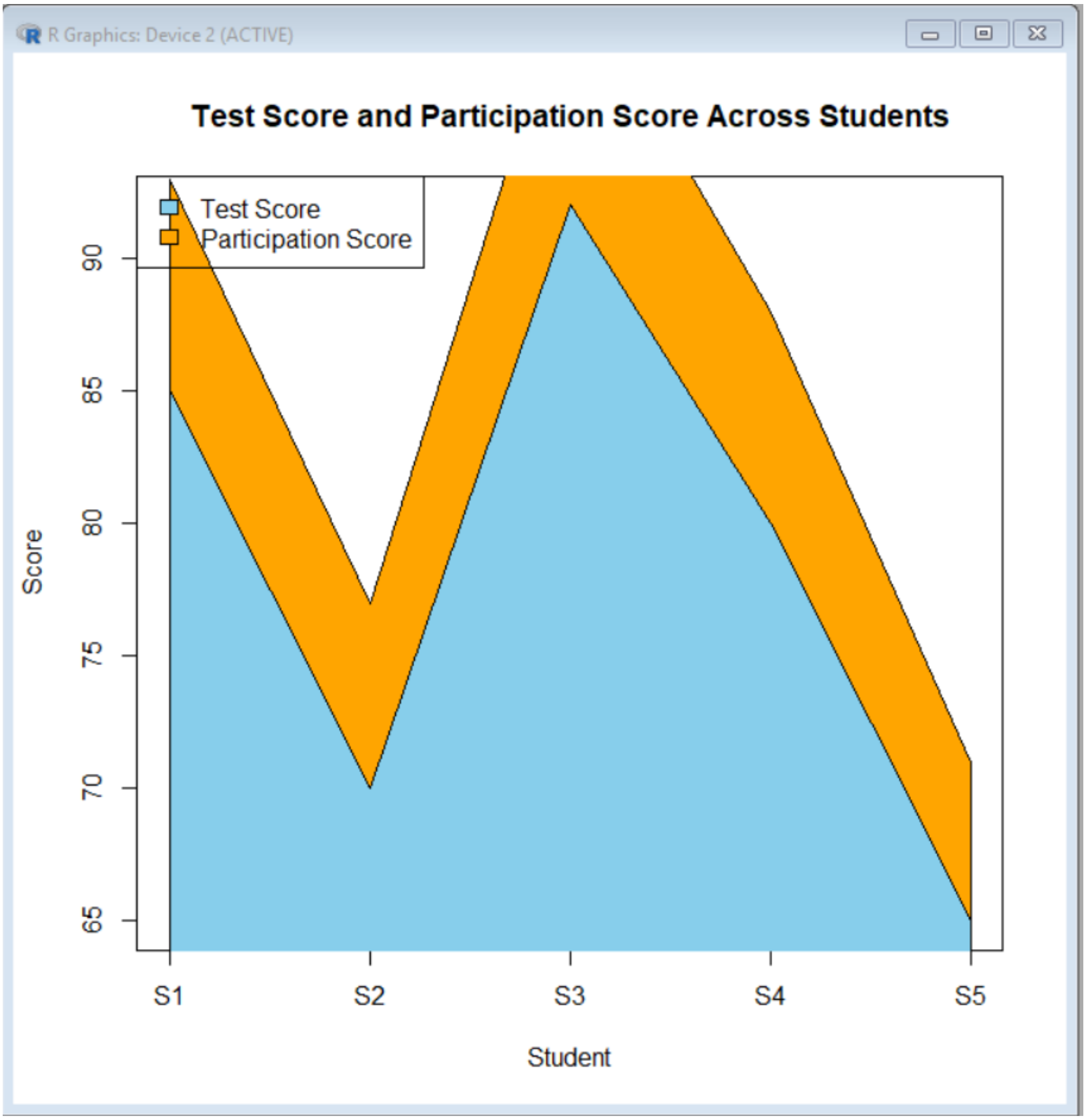
Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

```
# Create dataset
student <- data.frame(
  Student_ID = c("S1", "S2", "S3", "S4", "S5"),
  Age = c(19, 21, 20, 22, 23),
  Study_Hours = c(12, 8, 15, 10, 7),
  Attendance = c(90, 70, 95, 85, 60),
  Test_Score = c(85, 70, 92, 80, 65),
  Participation_Score = c(8, 7, 9, 8, 6)
)
```

```
# Create stacked area chart
test <- student$Test_Score
participation <- student$Participation_Score

plot(1:5, test,
    type = "n",
    xaxt = "n",
    xlab = "Student",
    ylab = "Score",
    main = "Test Score and Participation Score Across Students")

axis(1, at = 1:5, labels = student$Student_ID)

# Stacked areas
polygon(c(1:5, 5:1),
       c(rep(0, 5), rev(test)),
       col = "skyblue",
       border = "black")

polygon(c(1:5, 5:1),
       c(test, rev(test + participation)),
       col = "orange",
       border = "black")

# Legend
legend("topleft",
      legend = c("Test Score", "Participation Score"),
```

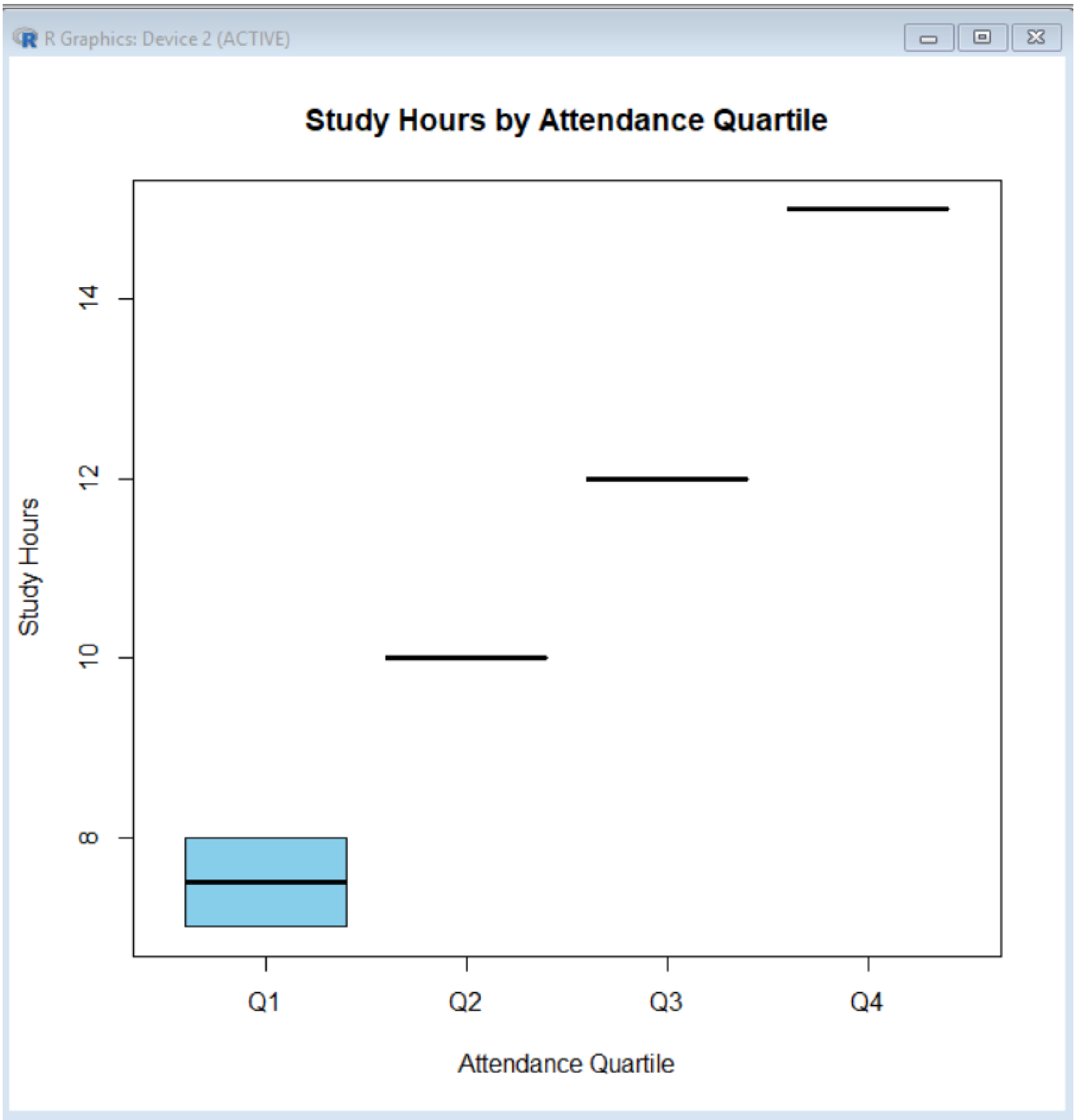
fill = c("skyblue", "orange"))



2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

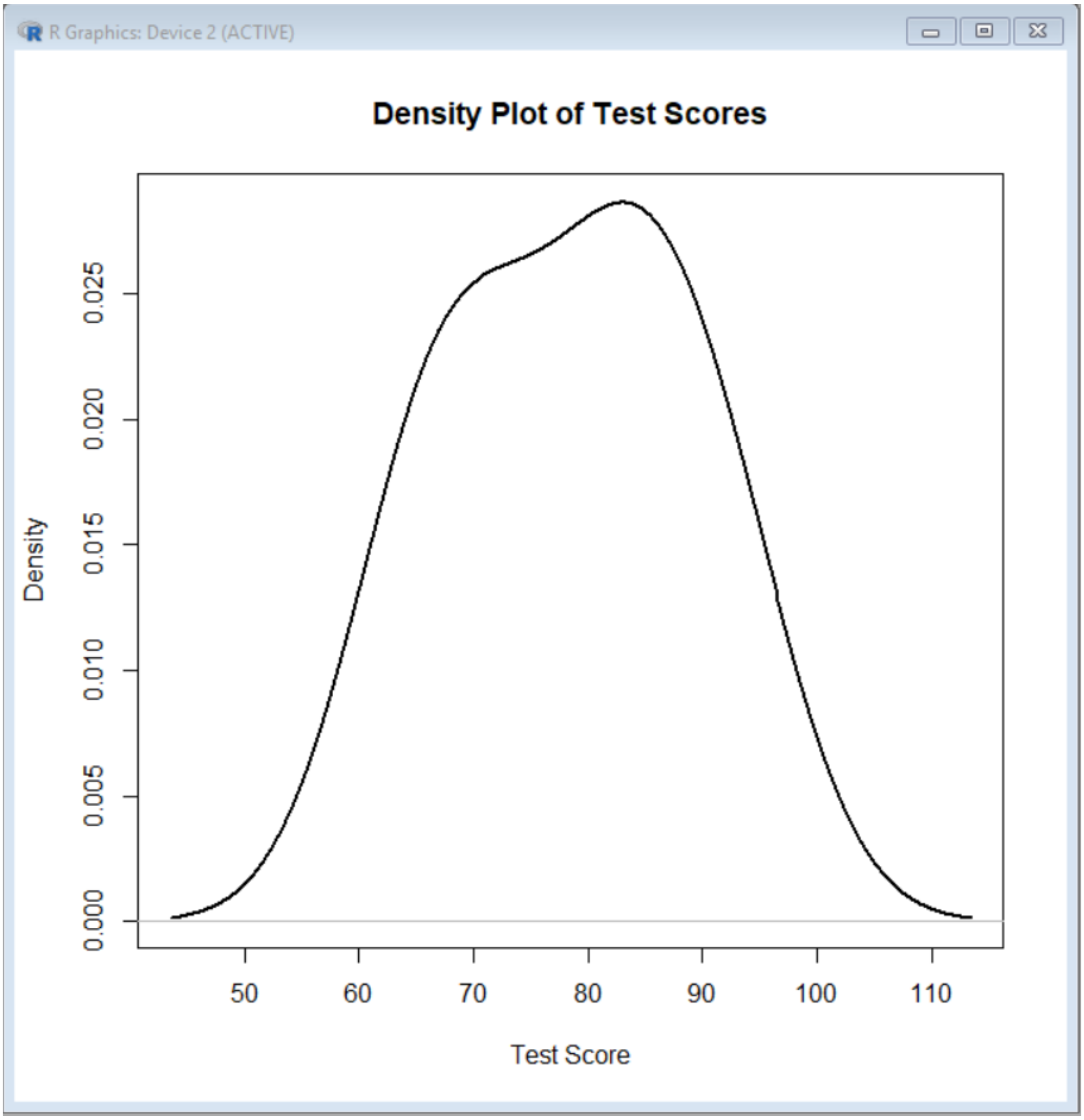
```
# Create Attendance quartiles
student$Attendance_Quartile <- cut(
  student$Attendance,
  breaks = quantile(student$Attendance,
```

```r
                   probs = c(0, 0.25, 0.5, 0.75, 1),
                   na.rm = TRUE),
  include.lowest = TRUE,
  labels = c("Q1", "Q2", "Q3", "Q4")
)

# Check the groups
print(student[, c("Student_ID",
                 "Attendance",
                 "Study_Hours",
                 "Attendance_Quartile")])

# Create boxplot
boxplot(Study_Hours ~ Attendance_Quartile,
        data = student,
        col = c("skyblue", "orange", "lightgreen", "pink"),
        main = "Study Hours by Attendance Quartile",
        xlab = "Attendance Quartile",
        ylab = "Study Hours")
```

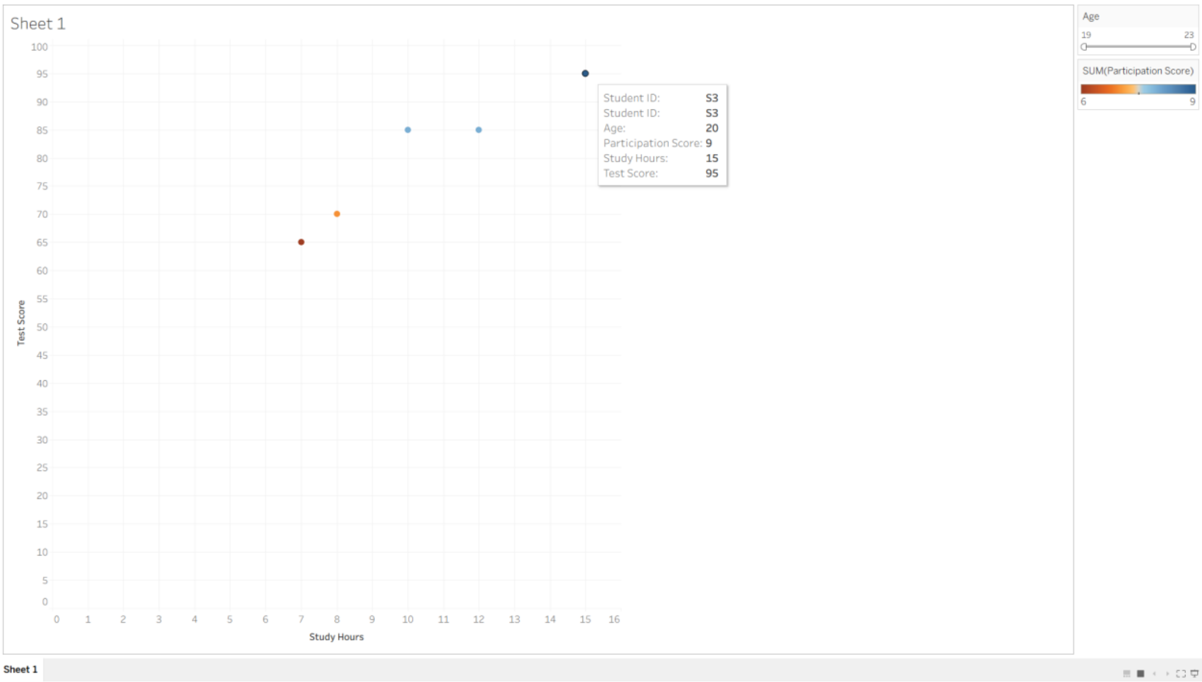


3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

```
# Density plot for Test Score
plot(density(student$Test_Score),
    main = "Density Plot of Test Scores",
    xlab = "Test Score",
    ylab = "Density",
    lwd = 2)
```



4. In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.



SET 30

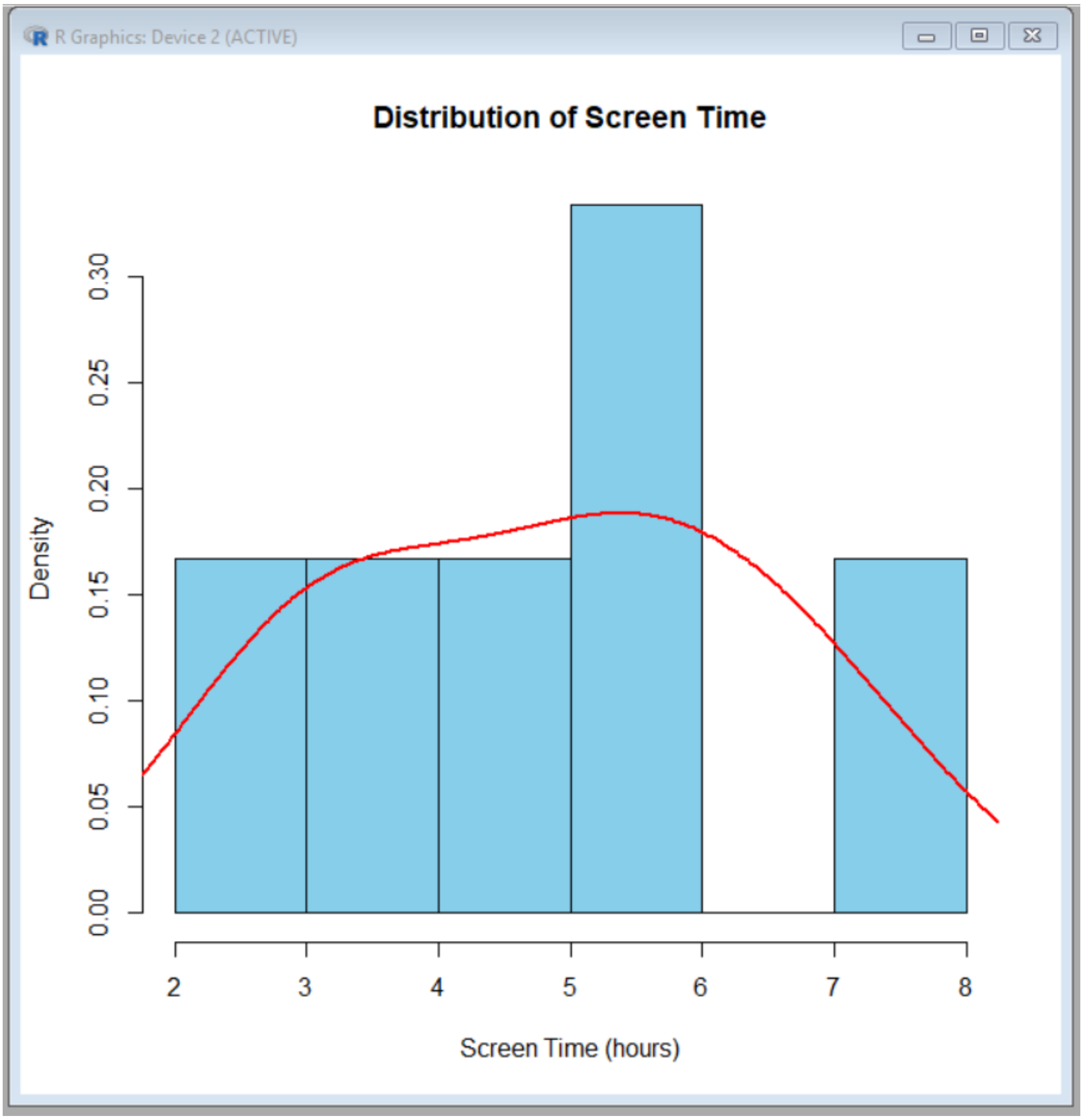
Dataset: Mobile_App_Usage.csv

User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

1. Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

```
hist(mobile$Screen_Time,
    breaks = 5,
    probability = TRUE,
    col = "skyblue",
    border = "black",
    main = "Distribution of Screen Time",
    xlab = "Screen Time (hours)",
```

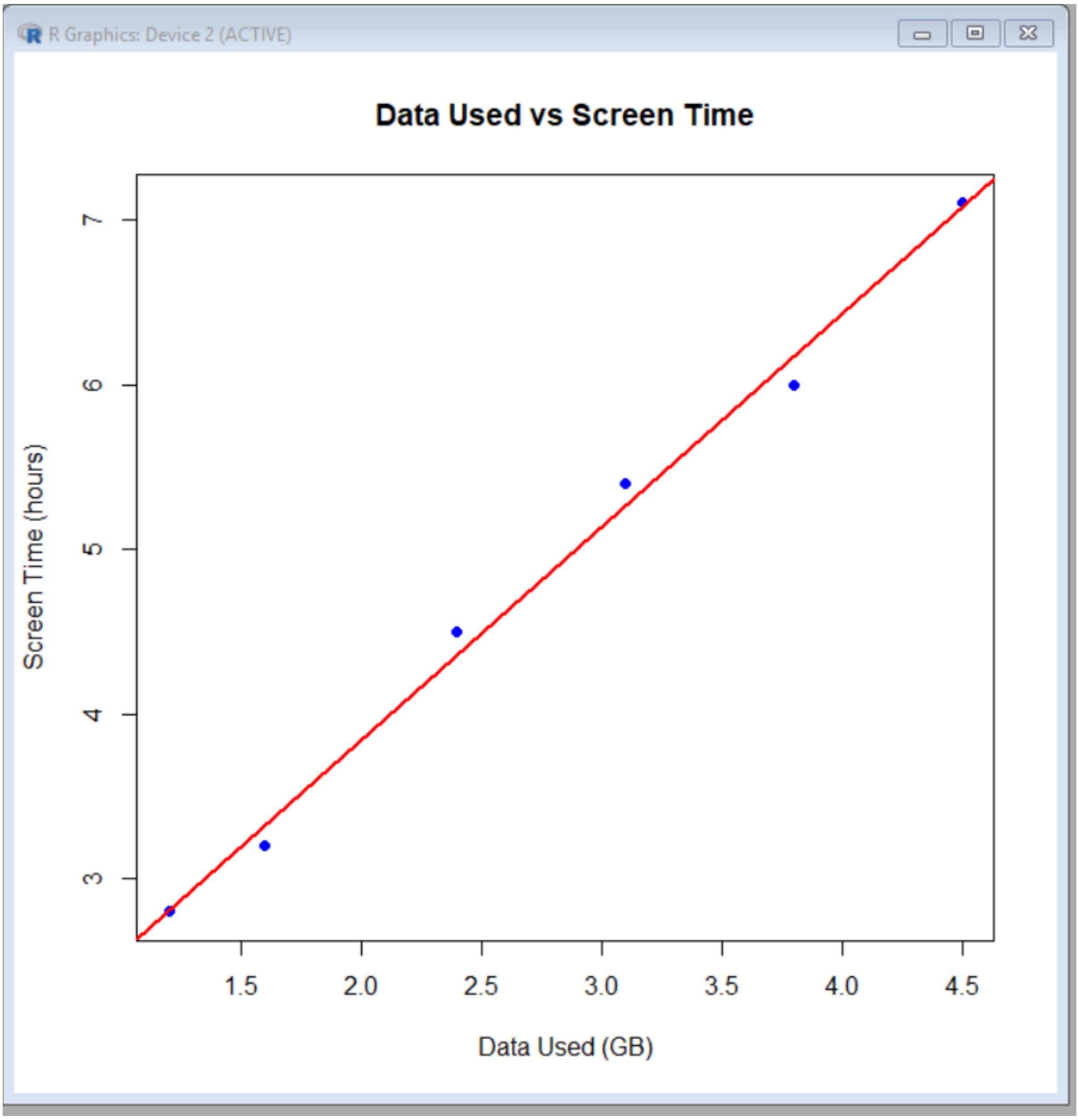
```
    ylab = "Density")

lines(density(mobile$Screen_Time),
      col = "red",
      lwd = 2)
```



2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

```
# Scatter plot

plot(mobile$Data_Used,
```

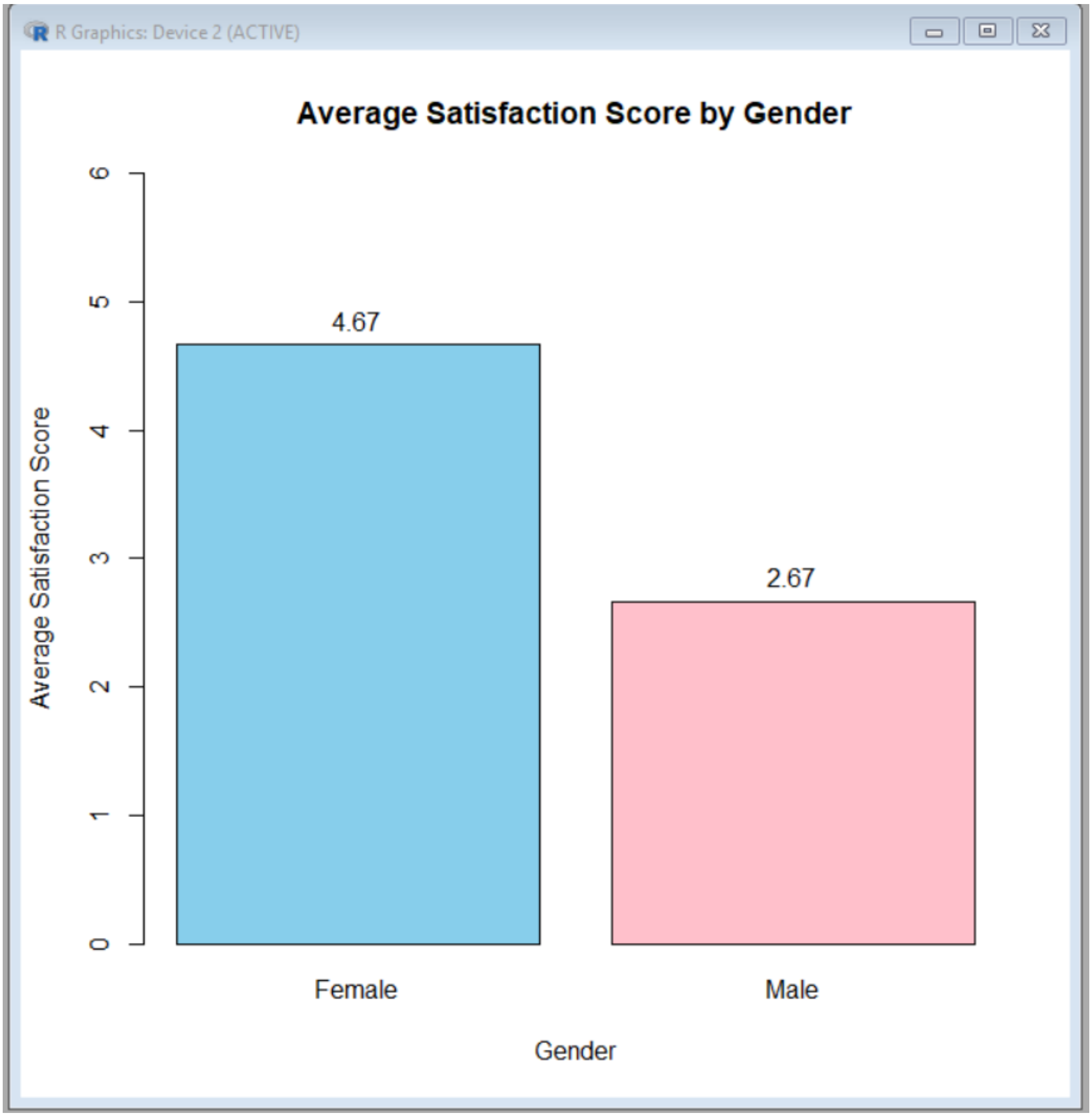
```
    mobile$Screen_Time,
    pch = 19,
    col = "blue",
    main = "Data Used vs Screen Time",
    xlab = "Data Used (GB)",
    ylab = "Screen Time (hours)")

# Add trend line
abline(lm(Screen_Time ~ Data_Used, data = mobile),
     col = "red",
     lwd = 2)

# Calculate correlation
correlation <- cor(mobile$Data_Used, mobile$Screen_Time)

print(correlation)
```



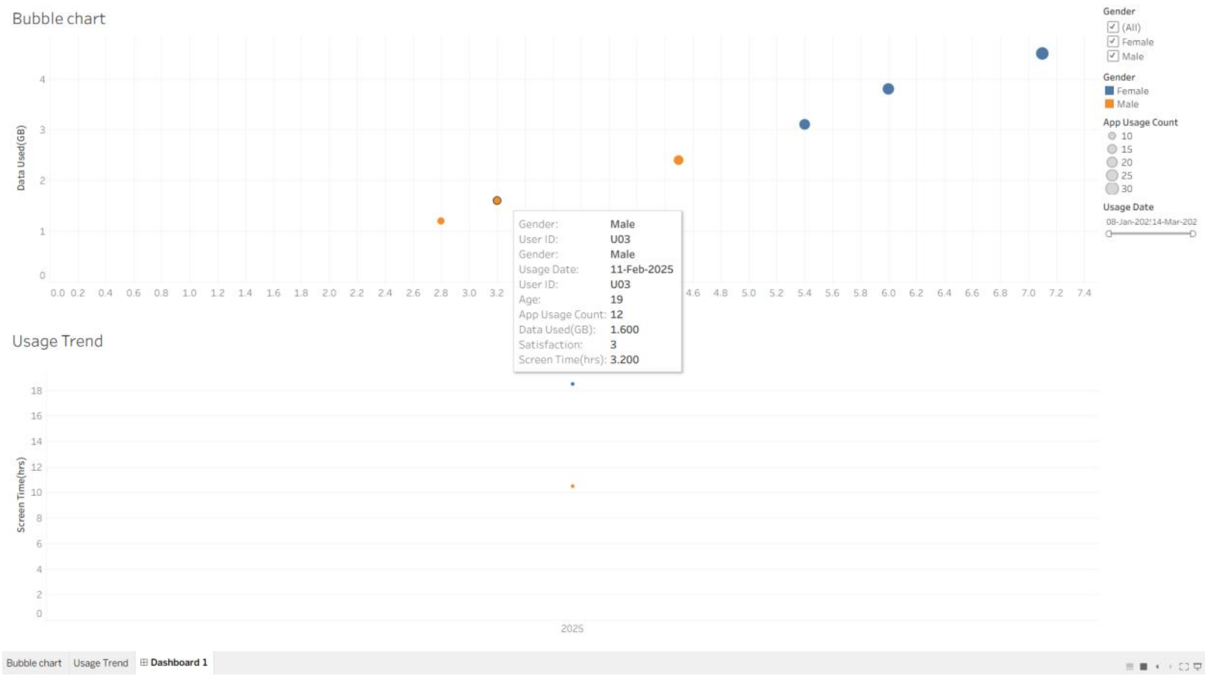
3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

```
# Calculate average Satisfaction by Gender

avg_satisfaction <- aggregate(Satisfaction ~ Gender,

                        data = mobile,

                        FUN = mean)


# Display averages

print(avg_satisfaction)
```

```r
# Create bar chart
bar <- barplot(avg_satisfaction$Satisfaction,
               names.arg = avg_satisfaction$Gender,
               col = c("skyblue", "pink"),
               main = "Average Satisfaction Score by Gender",
               xlab = "Gender",
               ylab = "Average Satisfaction Score",
               ylim = c(0, 6))

# Add value labels
text(x = bar,
     y = avg_satisfaction$Satisfaction,
     labels = round(avg_satisfaction$Satisfaction, 2),
     pos = 3)
```



4. Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used).Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.


Bubble chart
Data Used(GB)
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4 2.6 2.8 3.0 3.2 4.6 4.8 5.0 5.2 5.4 5.6 5.8 6.0 6.2 6.4 6.6 6.8 7.0 7.2 7.4
Gender: Male
User ID: U03
Gender: Male
Usage Date: 11-Feb-2025
User ID: U03
Age: 19
App Usage Count: 12
Data Used(GB): 1.600
Satisfaction: 3
Screen Time(hrs): 3.200
Gender
(All)
Female
Male
Gender
Female
Male
App Usage Count
10
15
20
25
30
Usage Date
08-Jan-202 14-Mar-202
Usage Trend
Screen Time(hrs)
2025
Bubble chart
Usage Trend
Dashboard 1